

Nikon Metrology X-ray and CT Inspection

Setting up a Circular CT scan (DRAFT)

Training Manual

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About This Manual

This manual applies to a typical machine with the model type given on the front cover. It contains a translation of the ORIGINAL Instructions for the safe use of this machine.

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Chapter 1: Background theory

Though the CT scan acquisition process is fairly straightforward, understanding of the background principles and how to apply them is crucial to getting the best quality data out of your Nikon Metrology system.

What are X-rays?

X-rays are a form of electromagnetic radiation: they are the same as radio waves and visible light but have a much higher energy.

Like light, they travel in straight lines and dissipate immediately when the source is switched off. Nikon Metrology X-ray systems do not use radioactive sources and therefore there is no ionising radiation emitted when the X-rays source is switched off.

X-rays are ionising radiation, which means they are harmful to living things (and some non-living things in high doses) - you should be aware of your local regulations on ionising radiation before using any X-ray system. With this said, Nikon Metrology X-ray systems are designed with radiation shielding and redundant safety features such that they present minimal risk to the operator when maintained and used properly. This manual will assume you have already familiarised yourself with your local rules and the safety features of your system; if not, now is the time to do so.

The energy of an electromagnetic wave can be described in a number of ways - you may be familiar with frequency (in Hertz) and wavelength (in metres). The energy of an X-ray is often described in electron-volts (eV), which is a unit of energy. Nikon Metrology X-ray sources produce X-ray energies in the hundreds of thousands of electron-volts: typically 30-450 keV.

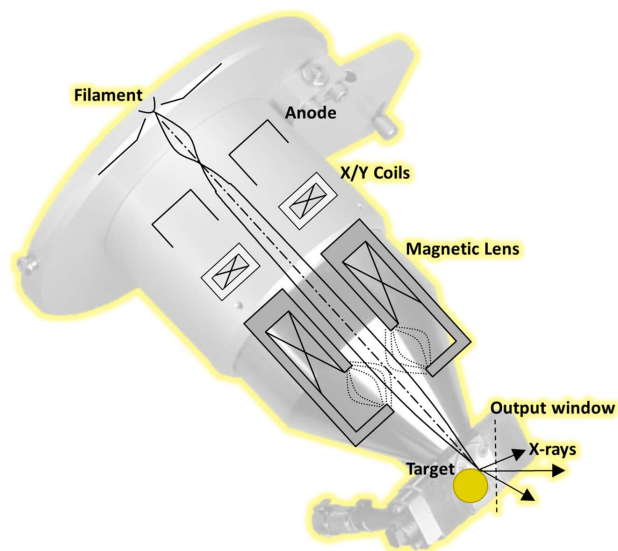
X-ray source types

Nikon Metrology offers a range of different X-ray sources, many of which can be interchanged by the operator.

Your system may have one or more targets for its X-ray source - in any case, the principles that apply to all of them are described below for the reflection target. It is recommended that you read the description of the reflection target, followed by any other targets your system includes (if any).

Reflection target

The reflection target is the most common type of X-ray source. All Nikon Metrology X-ray sources operate on the following principle: **aiming a very high energy, highly focussed electron beam at a metal target causes X-rays to be produced:**



1. A filament is heated until it glows. This hot filament releases electrons.
2. A very high voltage is applied between the filament (the cathode) and an anode. The anode attracts and accelerates the electrons into a high-energy beam.
3. A set of electromagnetic coils align and focus the electron beam to a very small focal spot.
4. The electron beam interacts with the target, producing X-rays.
5. The X-rays exit the source through the output window. The output window allows X-rays out of the source while sealing the high vacuum inside.

The processes that produce the X-rays (explained later) are very inefficient: over 99% of the power in the electron beam is converted to heat. The target is actively cooled, but this heat generation limits how much power can be deposited for a given focal spot size.

If too much power is focussed onto too small a focal spot, the target will be pitted (burnt) - to limit this, Inspect-X automatically defocuses the electron beam as power is increased to spread the heat out. As discussed later, this may affect image sharpness.

Rotating reflection target

The rotating reflection target replaces the static reflection target with a spinning disc.

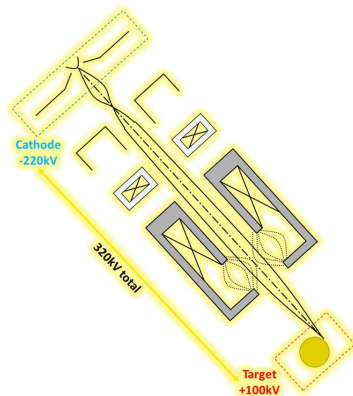


The target disc spins at high speed, with the electron beam landing on a tungsten track on the edge. This provides much better heat dissipation because any given piece of the target is only briefly heated by the electron beam before it is given a full rotation to cool down.

Compared to the static target, the rotating reflection target allows a significantly smaller focal spot size at higher power - providing scans which are faster and/or have less noise.

320 module

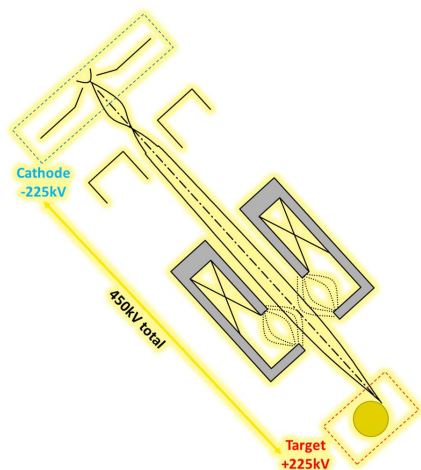
The 320 module is a target module with a built-in 100kV generator. When attached to a 225kV source it boosts the maximum beam energy to 320kV.



This allows better penetration through dense samples, plus a higher maximum power for faster scans or less noise.

450 static target

The 450 static target source applies +225kV to the target. This results in a total filament-to-target potential of 450kV.

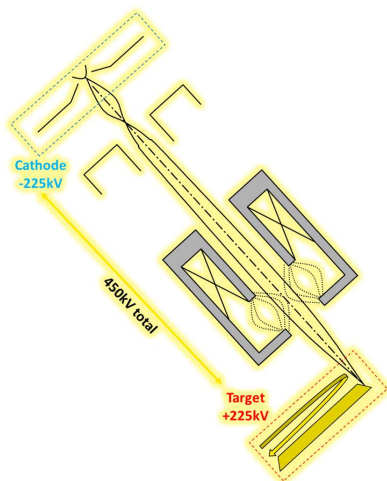


This provides significantly more penetration through dense samples.

450 rotating target

Like the 450 static target, the 450 rotating target source applies +225kV to the target. This results in a total filament-to-target potential of 450kV.

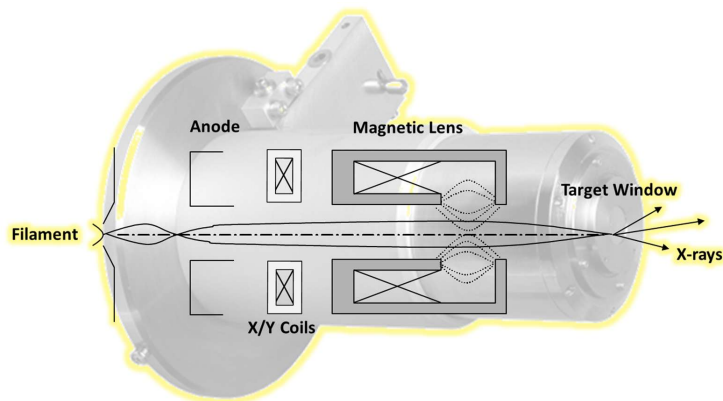
In addition, the target is a spinning disc which provides significantly better heat dissipation and therefore a significantly smaller focal spot size at high power (see *rotating reflection target*).



Compared to the 450 static target, the 450 rotating target allows a significantly smaller focal spot size at higher power - providing scans which are faster and/or have less noise.

Transmission target

The transmission target combines the target and output window into a single component: the target window.



The target is a thin film of Tungsten deposited on the inside face of the output window. This allows a significantly smaller focal spot than a reflection target and allows the sample to be placed much closer to the focal spot for higher geometric magnification.

X-ray production

When the high-energy electrons hit the target, two mechanisms act at the same time to produce X-rays. The explanations below of how these work are provided for your interest, but it's not necessary to memorise them - only their effects on the X-ray spectrum that is produced are important.

Bremsstrahlung Radiation

The atoms that make up the target have a positively charged nucleus. When an electron passes close by, it is attracted, and its path is deflected slightly towards the nucleus; the electron slows down and loses some energy. The energy lost by the electron is released as an X-ray.

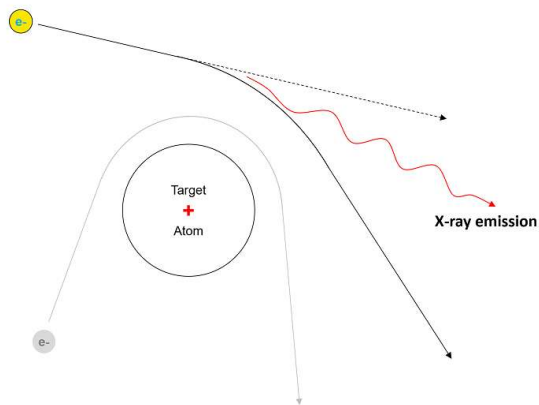


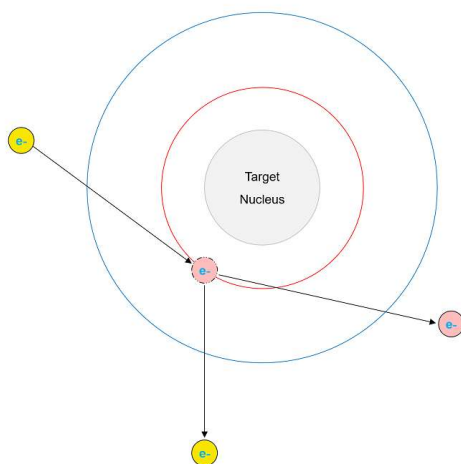
Figure 1. Electron interaction for generating a Bremsstrahlung X-ray

Some electrons are deflected more than others: those that are deflected a lot release high-energy X-rays and conversely, those that are only slightly deflected don't lose much energy so release low-energy X-rays. Therefore, **a continuous spectrum of X-rays is produced.**

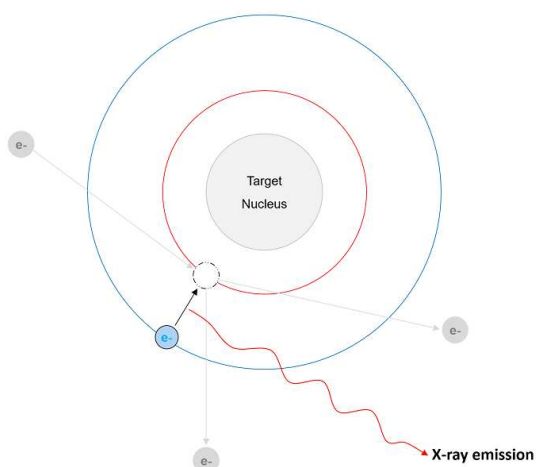
Characteristic Emission

Some of the high-energy electrons collide with the electrons surrounding the atoms in the target, knocking them free. This leaves a hole behind; another electron in the atom "falls down" to fill the hole. This transition from a higher- to lower-energy state releases energy in the form of an X-ray.

1: High-energy electron collides with electron in target atom



2: Electron already in the atom loses energy to fill the hole, releasing an X-ray

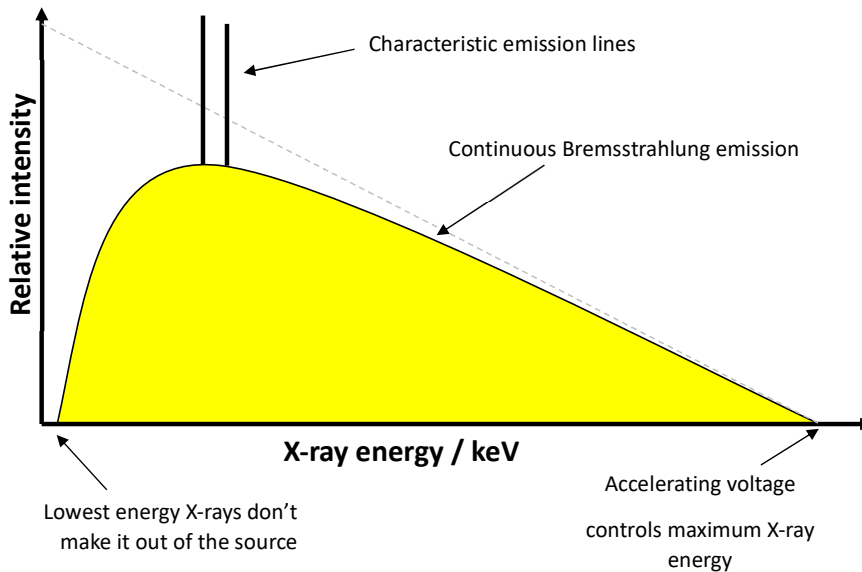


Because there are only very specific energy levels the electrons can be in, there are therefore only certain transitions possible - this means that **only X-rays of specific energies are produced**. The energies of this X-ray emission are characteristic of the metal that the target is made from.

The X-ray spectrum

This combination of Bremsstrahlung radiation and characteristic emission generates the X-ray spectrum.

Simplified diagram of an X-ray spectrum:



The spectrum contains X-rays with energies up to (but not more than) the accelerating voltage - for example, a 200kV accelerating voltage would result in X-rays **up to** 200keV.

The majority of X-rays are of much lower energy than the accelerating voltage - this is important to bear in mind as it is very important when setting up CT scans. This will be discussed in more detail later.

Multi-metal reflection target

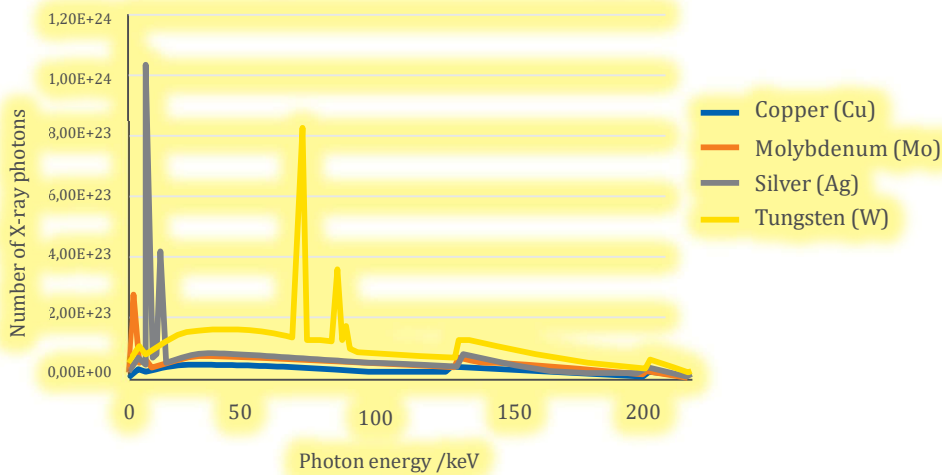
The standard reflection target is made from Tungsten (W) because of its intense characteristic emission around 60keV. At lower accelerating voltages, Tungsten no longer provides this intense characteristic emission.

Different metals can be used as X-ray targets: those with characteristic emission at lower energies than Tungsten will provide intense emission at those lower energies. Although the X-ray beam may not be any brighter when using different metals (it may actually be dimmer), there will be more low-energy X-rays resulting in better contrast.

As an option for the 225kV reflection target, the multi-metal target offers four metals which can be indexed without breaking vacuum. Each metal emits a range of characteristic X-ray energies; these **approximate** ranges are:

- Tungsten (W) (58-69 keV)
- Silver (Ag) (22-26 keV)
- Molybdenum (Mo) (18-20 keV)
- Copper (Cu) (8-9 keV)

X-ray radiation spectrum for different target materials



The different metals may be advantageous for samples that are very light and provide very little contrast. For general use above 50kV, Tungsten is recommended.

Casting a shadow

When light shines upon an object, the object will absorb some of the light and cast a shadow. The same occurs with X-rays: X-ray images (*radiographs*) are just shadows!

Most objects are opaque and therefore light won't pass through them: the shadow that is cast by visible light is usually a silhouette. X-rays, however, will penetrate through any material so leave a shadow of not only the outside shape but all the internal features too.



A sample placed in the X-ray beam casts a shadow

Attenuation

The attenuation of an X-ray beam (the amount of X-rays which were absorbed) depends on how much material the beam has to pass through; denser and/or thicker objects will cast darker shadows.

Commented [JF1]: ALSO: need to add section saying lower energy X-rays are absorbed easier than higher energy

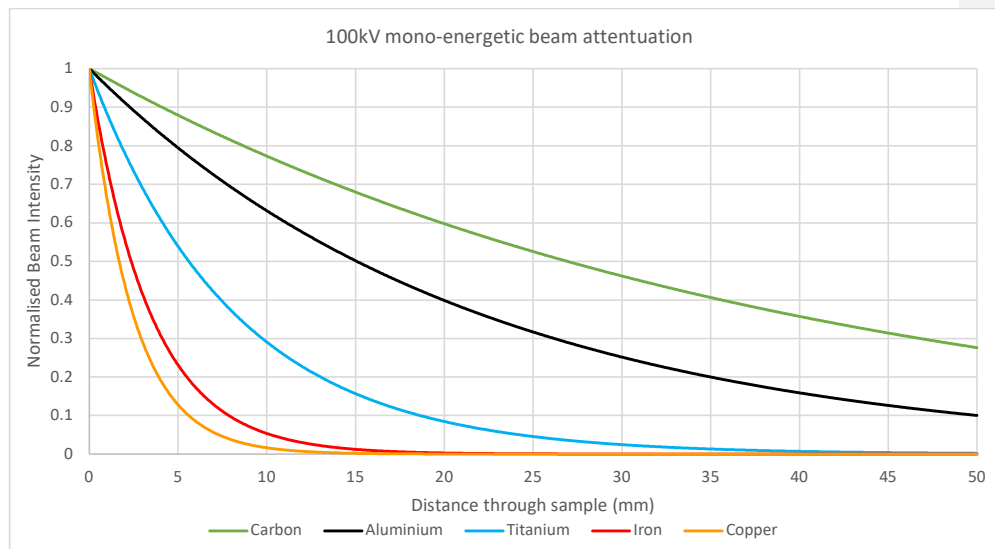


Figure 2. X-ray attenuation through different materials. A poly-energetic beam (e.g. 215keV (peak) with 1.5mm Cu filter) can be treated as a single, lower energy mono-energetic beam for the purposes of attenuation, in this case ~100keV. [ref: [X-Ray Mass Attenuation Coefficients | NIST](#)]

X-rays are absorbed as an exponential decay. This means that, for example, if one thickness of material absorbs half of the X-rays then the same thickness again will absorb half of what remains - transmitting a quarter of the original beam. This follows as 1/8th, 1/16th, etc. Although some X-rays will technically penetrate through any thickness of any material, there is always a practical limit to penetration where there are too few X-rays remaining to detect.

The density of electrons in the sample material affects how rapidly X-rays are attenuated; samples with a higher atomic (Z) number or higher density will absorb more X-rays.

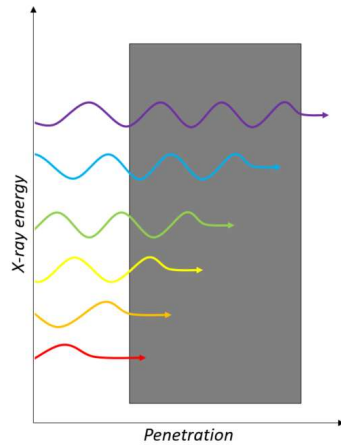


Figure 3. High-energy X-rays penetrate further through the same material than low-energy X-rays.

One behaviour of the X-ray beam that is not immediately obvious is that since the source generates a beam consisting of multiple energies they will attenuate at different rates. This means that the lower energies of the beam attenuate faster than the higher energies, resulting in a 'hardening' of the beam as it travels through a given material (as illustrated above).

X-ray conditions

X-ray conditions explained

Inspect-X provides three main parameters which control the X-ray source: **Beam Energy (kV)**, **Beam Current (μA)** and **Power (W)**.

- Beam energy controls the maximum X-ray energy in the X-ray beam. Higher beam energy provides more high-energy X-rays resulting in more penetration.
- Beam current controls the quantity of X-rays being generated. Higher current results in a brighter image.
- Power is the product of beam energy and beam current:

$$\text{Power} = \text{Beam Energy} \times \text{Beam Current}$$

The power control does not directly control the source; adjusting the power indirectly adjusts the beam current.

Because beam current and power depend on each other, only one of them needs to be directly controlled. **The focal spot size of microfocus sources depends directly on power**, so the power control is most often used because it lets us easily estimate the focal spot size.

X-ray source focal spot size

Each source type has a different range of possible focus spot sizes, which is a result of the optics within the source.

To not pit the target, all Nikon Metrology microfocus X-ray sources defocus the X-ray focal spot size roughly linearly with power.

In general Nikon Metrology *static target* sources defocus at approximately $1\mu\text{m}/\text{W}$. For one of these sources this means that at 50W you can expect the focal spot size to be around $50\mu\text{m}$.

Nikon Metrology *rotating target* sources on the other hand defocus at approximately $1\mu\text{m}/3\text{W}$, which means for a power of 150W you would still achieve a focal spot size of $50\mu\text{m}$.

In the table below, the best focal spot size is listed with the power it is achieved at. This means that there is no benefit to using a power below this as it **does not improve the resolution further**.

The maximum power of the X-ray source is also listed below.

Source type	Best focal spot size @ best spot power limit	Maximum focal spot size @ maximum source power
180kV transmission	$1\mu\text{m}$ @ 3W	$20\mu\text{m}$ @ 20W
225kV static reflection	$3\mu\text{m}$ @ 7W	$225\mu\text{m}$ @ 225W
225kV rotating reflection	$10\mu\text{m}$ @ 30W	$160\mu\text{m}$ @ 450W
320kV (static) reflection	$30\mu\text{m}$ @ 20W	$320\mu\text{m}$ @ 320W
450kV static reflection	$80\mu\text{m}$ @ 50W	$320\mu\text{m}$ @ 450W
450kV rotating reflection	$80\mu\text{m}$ @ 100W	$120\mu\text{m}$ @ 450W

X-ray beam filters

Most X-ray sources produce a whole spectrum of X-ray energies, known as a polychromatic beam. This means that if the X-ray source is set to, for example, 200kV then although there will be some emission of 200keV X-rays the vast majority of X-rays will be of lower energy (as seen previously).

This polychromatic spectrum causes two effects:

1. Reduced penetration

Because most X-rays being produced are of lower energies, they do not contribute to penetration through the sample but do light up the detector - this results in a limit to penetration because the detector becomes saturated if we try to increase flux (brightness). If these lower energy X-rays can be removed from the beam, the resulting beam will naturally be much darker; however when the flux is increased back to the previous level (by increasing power or exposure) the resulting beam will have much greater penetration because it contains a much higher proportion of high-energy X-rays - it is "harder" and has a higher mean energy.

2. Beam hardening

Beam hardening is a very common artefact in CT datasets and is a result of the lower energy X-rays in the beam being absorbed much more quickly than the higher energy part of the beam.

This behaviour produces a bright-edge artefact in dataset as we can't tell where the X-rays were absorbed in the sample, and there is no beam-energy information with detectors commonly used. This means that the first thickness of a material has a higher '**effective attenuation coefficient**' which is shown as the bright edge to the sample.

This can be corrected in two ways:

a. Filter the X-ray beam

By adding a filter to the X-ray source, we preferentially remove lower energy X-rays and therefore make the X-ray beam closer to a mono-energetic source and as a result minimise the artefact as part of the scan (which gives the best results for data quality).

To illustrate this, we will consider an aluminium disk with a diameter of 40mm; and consider the attenuation through a chord of 4.5mm. **It's the difference in attenuation between the full, and glancing/small pathlength that generates the beam hardening artefacts.**

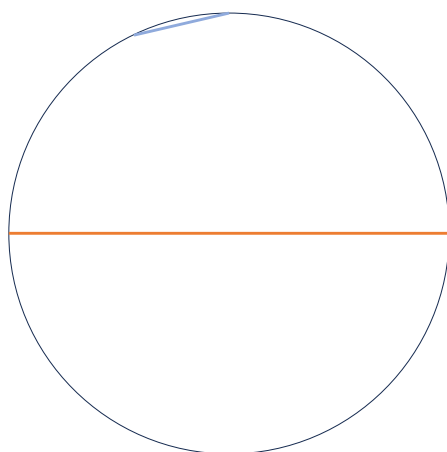


Figure X. Disk of 40mm diameter (orange) and considering a chord of 4.5mm (blue). Chord length chosen arbitrarily (but the more tangential the chord chosen, the more extreme the difference between chord and diameter effective attenuation that will be observed).

Using SpekPy (<https://spekpy.smile.ki.se/>) we can calculate the **effective attenuation coefficient** for the chord and diameter using X-ray beams with different filters attached. We find that as we **increase the filtration, the attenuation coefficients of the chord and diameter become more similar**:

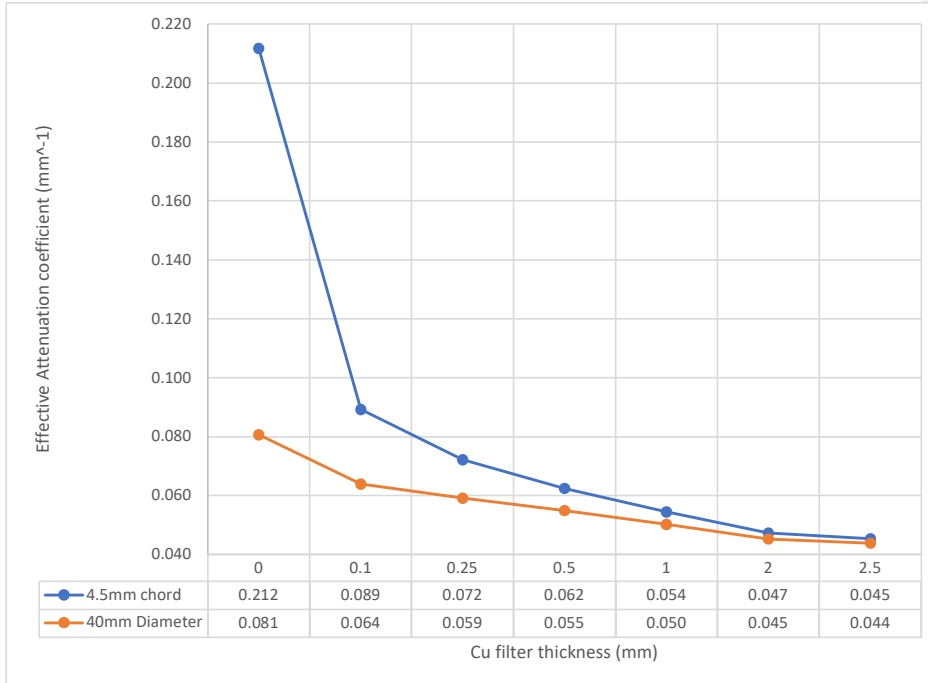


Figure XX. Attenuation coefficients of the chord and diameter of the disk. A higher value indicates that a higher fraction of the original X-ray beam was absorbed along that pathlength.

The result of this is immediately clear in the reconstructed CT datasets:

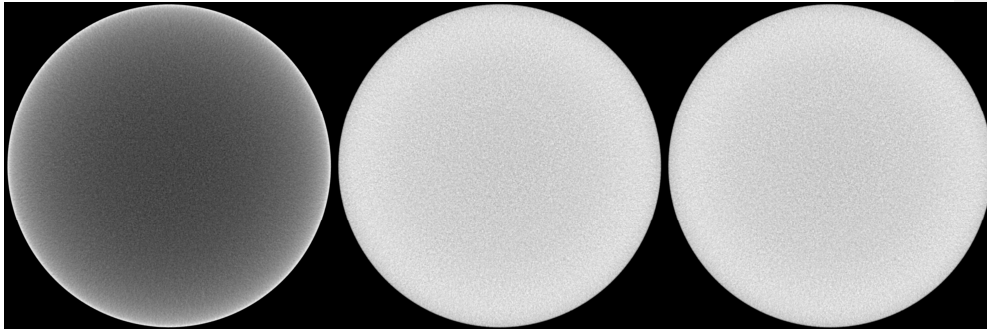


Figure XXX. Reconstructed slices of the Aluminium disk. Left to right: 0mm, 1mm Cu, 2.5mm Cu filtration. The tone curve has been adjusted to highlight the beam hardening present in the dataset.

And finally, we can confirm that the theoretical attenuations correlate with the greyvalue difference from the edge to the centre of the disk (which is the measure of the beam hardening artefact strength) by plotting them together:

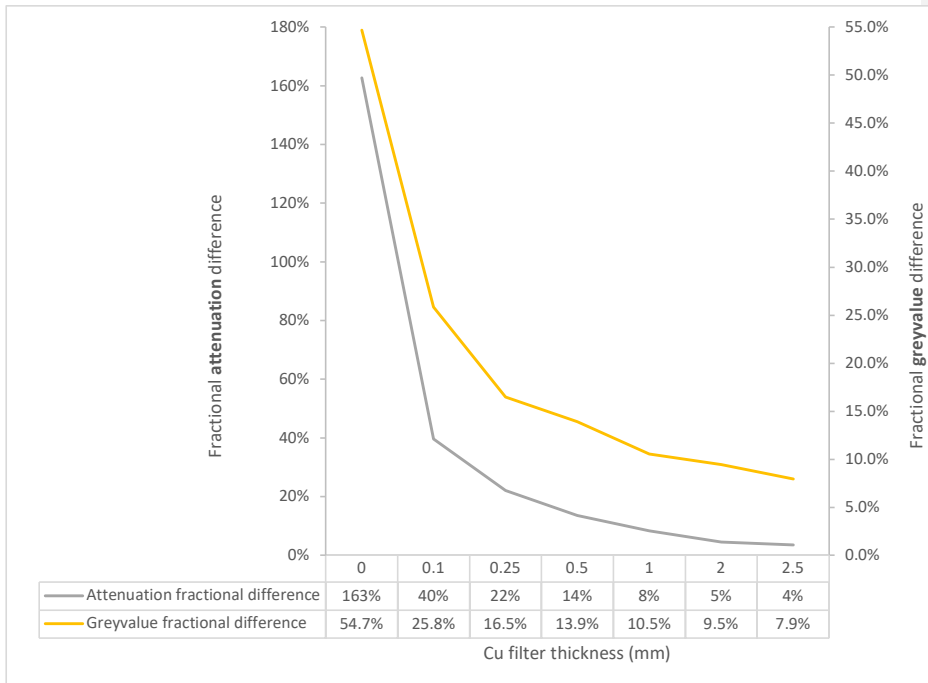


Figure XXXX. Fractional differences between the chord and diameter attenuations, and edge-to-centre greyvalues plotted together.

We can see from figure XXXX that we get the majority of the correction from a very small amount of the filter (e.g. the initial 0.1mm thickness gives us 62% of the correction of the 2.5mm filter); and that we reach a point of diminishing returns from increasingly thick filters.

In general, the effective attenuation coefficient difference is more extreme for more dense materials.

This means for low density samples (e.g. carbon fibre) 0.1mm Cu filter could almost eliminate all the beam hardening, while high density samples (e.g. steel) would require much higher filters to reach similar levels of greyvalue homogeneity.

b. Beam hardening correction in reconstruction

If scan conditions cannot eliminate the beam hardening (i.e. when it's not possible to add enough beam filtration, or fast scan time is a priority), it is possible to correct it during reconstruction.

It must be noted that software correction can only optimally correct for one material at a time, so multi-material samples may have beam hardening present even when the reconstruction correction is applied.

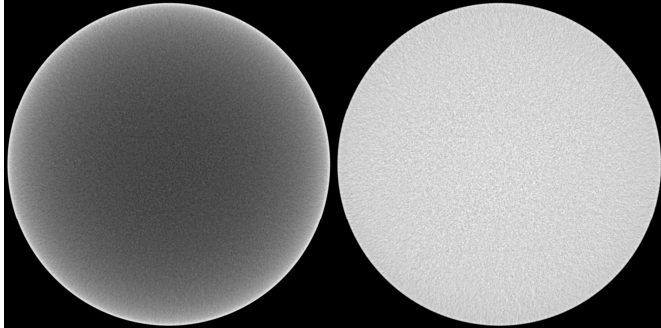


Figure XXXXX. Aluminium disk scan with no beam filtration. Left: raw dataset. Right: dataset with beam hardening correction applied.

While the best possible scan results in general will result from using the physical beam filtration, the software correction has reduced the edge-to-centre greyvalue difference from 54.7% to 7.9% which is on par with the 2.5mm Cu scan for this single material sample (which had adequate penetration even without any filtration).

Using this route would have produced very similar results to (but in a fraction of the time of) the higher filtration scan.

X-ray scattering

This section covers X-ray scattering. It isn't essential to be familiar with this when setting up a CT scan, it is useful to have some understanding of it as a part of a foundation of knowledge around X-rays and their interactions with matter.

X-ray scatter (as well as its general attenuation in matter) is caused by the X-ray's interaction with the sample's atomic electrons.

The first scatter mode is **photoelectric absorption**. This is where an incident X-ray (with energy, E_0) collides with an inner shell electron and ejects it. This causes a higher shell electron to fill this gap in energy level and emits a characteristic photon in a random direction. The ejected electron may recombine with a different atom but is much more likely to lose energy via collisions (as heat to the rest of the material).

The second mode is **elastic scattering** (also called Rayleigh scattering), where an X-ray collides with an electron, but rather than eject it from the atom, it simply releases the photon (still with energy E_0) in a random direction.

The last mode is **inelastic scattering** (also called Compton scattering) where the X-ray collides with a weakly bound outer-shell electron and ejects it. The photon is also released, but now has a lower amount of energy. This also makes the resultant X-ray more likely to interact via scattering or photoelectric absorption.

In general, the lower the energy X-ray, the more likely it is to interact with matter (both via scattering and simply attenuation). The graph below illustrates the interaction cross section (which is a measure of how likely the event is) versus photon energy eV.

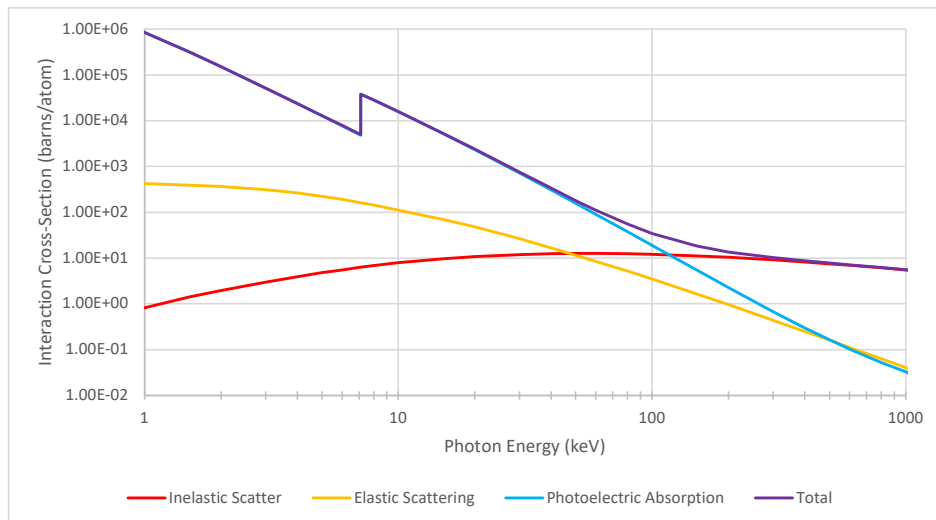


Figure 5. Modes of X-ray scattering for Iron, Fe. Typical X-ray sources generate up to 450keV photons. Photoelectric Absorption dominates at low energies, while Inelastic Scattering dominates at higher energies. Ref: [XCOM: Element Options \(nist.gov\)](https://xcom.nist.gov/)

This means, if we want to reduce scatter, we need to either:

- Increase the average energy of the X-ray spectrum, to reduce the likelihood of scattering interactions.
- Collimate the X-ray source and detector so you don't accept the scattered photons (such as with the CLDA (Curved Linear Detector Array)).

Like X-ray attenuation, overall scattering increases as the density of the material increases. Below is a comparison of the total scattering exhibited by Carbon, Aluminium, and Iron. The values of scattering are not critical to remember – only the relationship between material density and scattering is important.

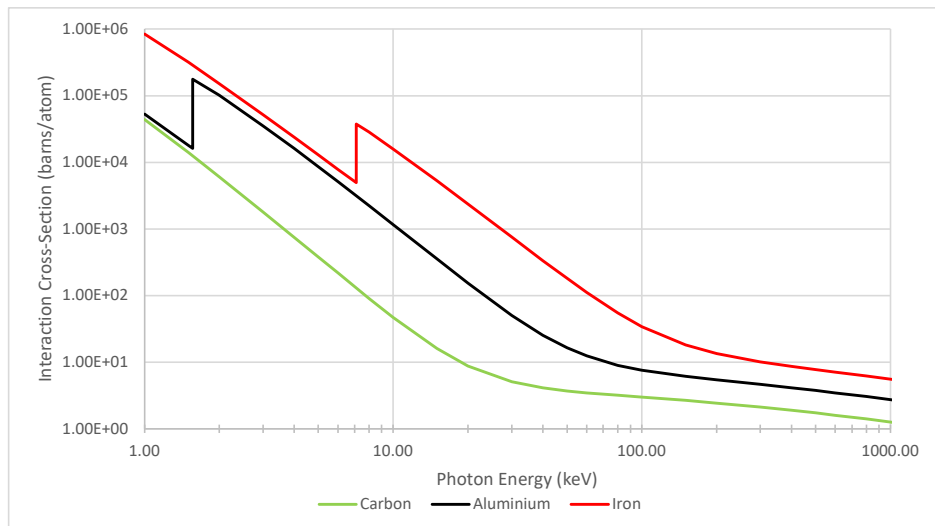


Figure 6. Total scattering cross sections for Carbon, Aluminium, and Iron. Higher density materials generate more scattering. Typical X-ray sources generate up to 450keV photons.

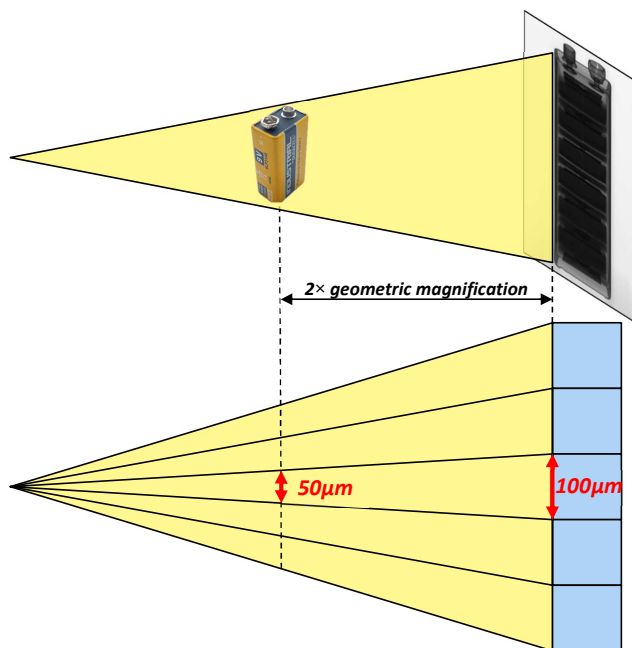
As an illustration for this effect, if we take the same theoretical conditions as before – 215keV (peak) with 1.5mm Cu filter – we can see that the 100keV effective beam energy the elements have a scattering cross section of: Carbon – 3.02, Aluminium – 7.64, and Iron – 30.98 barns/atom.

These effects are what make dense materials challenging to get good CT data from – they are harder to penetrate and generate far more scattering effects.

X-ray imaging

Effective pixel size

The X-ray beam starts as a small point in the source, spreading as a cone towards the detector. If an object is halfway between the source and detector, the image that is projected onto the detector will be twice as large as the object itself: the **geometric magnification** factor is $2\times$. Geometric magnification also occurs with shadows from visible light - try moving your hand closer to and farther from a small, bright light and watch the shadow.



If the pixels on the detector are, for example, $100\mu\text{m}$ in pitch then at $2\times$ magnification they represent $50\mu\text{m}$ within the sample: the **effective pixel size** is $50\mu\text{m}$.

A CT dataset consists of 3D grid of **voxels** (volume elements, i.e. 3D pixels); when the scan is reconstructed, the effective pixel size becomes the **voxel size**.

Image resolution

It's not uncommon to quote voxel size as the resolution of the scan but this is misleading - if you have $10\mu\text{m}$ voxels that are all blurry then you won't be able to resolve $10\mu\text{m}$ details and therefore the resolution is worse than $10\mu\text{m}$.

The other main limitation to resolution is the focal spot size of the X-ray source. Especially at high magnifications, the focal spot size can affect resolution greatly.

$$U = \frac{1}{M} \sqrt{p^2 + s^2(M-1)^2}$$

The equation for Geometric unsharpness, where:

U: Geometric unsharpness
M: Geometric magnification
p: Detector pixel size
s: X-ray source focal spot size

Geometric unsharpness defines the overall blurring caused by the focal spot size, detector pixel size and geometric magnification. Though it's not practical to calculate it in your head, at moderate to high geometric magnification we can use the following approximation:

If the X-ray source focal spot size is no larger than the effective pixel size, blurring caused by the focal spot size is minimal.

By following this rule, the resolution is limited primarily by magnification.

While the 1µm/W or 1µm/3W (depending on whether you're using a static or rotating source) is sufficient, the equations listed below can be used to calculate a more accurate power for the rule.

These are only valid above the best focal spot size (which are also listed for easy reference), for example, the equation for the 225kV static reflection source is only valid if the effective pixel size is greater than 3µm (since if it is lower a power of 7W can be used regardless of effective pixel size).

In the equation - E_p represents the effective pixel size.

Source type	Ideal max power (W)	Best focal spot size	Max power
180kV transmission	$\frac{17E_p + 40}{19}$	1µm @ 3W	20W
225kV static reflection	$\frac{109E_p + 450}{111}$	3µm @ 7W	225W
225kV rotating reflection	$\frac{14E_p + 10}{5}$	10µm @ 30W	450W
320kV (static) reflection	$\frac{30E_p - 320}{29}$	30µm @ 20W	320W
450kV static reflection	$\frac{5E_p - 250}{3}$	80µm @ 50W	450W
450kV rotating reflection	$\frac{35E_p - 2400}{4}$	80µm @ 100W	450W

Using a smaller spot size will improve the image sharpness but may not yield an increase in data quality due to the higher levels of noise present within the data (as a result of using less flux, see below).

Image noise

All images have some amount of noise in them. Most of the noise in an X-ray image is caused by the fact that there are only so many X-rays that hit the detector in a given time - this quantity of X-rays in an amount of time is called **flux**.

CT is particularly sensitive to noise, so it's important to reduce noise as much as reasonably possible.

With a photo camera, noise can be reduced by 3 main methods:

1. Increasing the amount of light available
2. Increasing the exposure time
3. Digitally reducing noise in post-processing

These are analogous to the following:

1. Increasing the X-ray power
2. Increasing the exposure time
3. Digitally reducing noise during reconstruction

We therefore want to use as high a power and as long an exposure as possible, given two limitations:

- If the power is too high, this will bring the focal spot size larger than the effective pixel size and cause blurring.
- The longer the exposure is, the more time the scan will take to acquire.

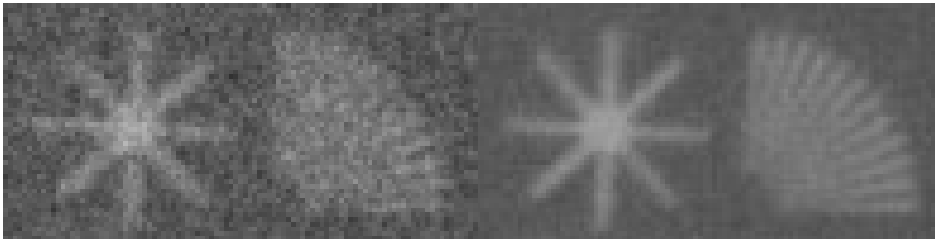


Figure 7. Left: 1 second exposure, Right: 16 second exposure

Image statistics and detector limitations

The **image statistics** next to the histogram summarise the greyvalues in the image. Of particular interest are the **min** and **max** values - these represent the greyvalues of the very darkest and brightest pixels on screen respectively.

Regardless of detector type, Inspect-X reads out greyvalues as 16-bit integers, i.e. whole numbers from 0 - 65,535 where 0 represents no signal and 65,535 represents the brightest illumination possible.

All detectors have a **saturation point** - this is a greyvalue above which the image is so bright that pixels may no longer respond reliably. For Varex PaxScan and CMOS detectors this is around 55,000 and for Varex XRD detectors this is 60,000.

Some detectors may have a **gain** option. This alters the sensitivity of the detector both to X-rays and noise - each step of gain will result in an image that is roughly half or twice as bright. Adjusting the gain will not directly affect the signal-to-noise ratio but allows you to make the best use of the full greyvalue range of the detector.

Gain is not to be confused with Digital Gain. Digital Gain provides no advantage for CT and should be left at 1 (disabled).

Shading corrections: Background

The construction of Digital Flat-Panel Detectors causes a variation in sensitivity across the detector which is most visible as stripes across the image. This is normal but must be corrected before a CT scan can be acquired. By acquiring images with no sample in view, Inspect-X generates a **shading correction** to produce a consistent, flat image.

Because the detector's response changes slightly with changes in X-ray conditions, imaging conditions and time, a new shading correction is required for every different set of conditions, or if too much time (typically a few hours to a day) has elapsed. If a valid shading correction exists, Inspect-X will use it automatically. Otherwise, a new shading correction will be automatically generated.

A single shading correction is applied to every projection image in the CT scan, so it is critical that there is minimal noise in the shading correction reference images. Shading correction reference images are therefore usually an average of multiple frames over a period of time to reduce noise.

The simplest form of shading correction uses 2 images: a white (gain) image and black (offset) image. This is sufficient for most scans across most detector types, however Inspect-X allows more than 2 images to be acquired - in this case grey image(s) at lower current will be acquired to correct for a non-linear sensitivity to X-rays.

Poor shading correction setup can lead to ring artifacts appearing in the CT reconstruction.

Projections

During the CT scan, **projection images** are acquired as the sample rotates. Each of these is a single radiograph (2D X-ray image) from a specific angle of rotation.

For each scan there is an optimum number of projection images, depending on the number of pixels in use across the detector. Increasing the number of projections above this will not significantly improve image quality, however the image quality will degrade as the number of projections is reduced: fringes will start to appear from the edges of the reconstruction.

As a rule of thumb, the number of projections can be reduced by approximately 1/3 without significant degradation in quality - but this should only be done as a last resort if there are no other options left to reduce the scan time.

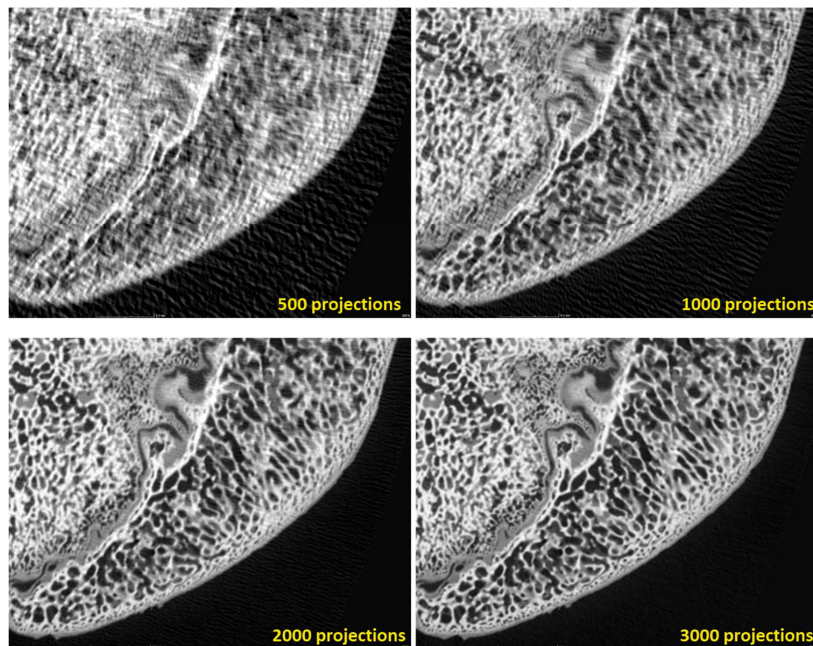


Figure 8. CT scan of a bone using different numbers of projections during data reconstruction.

Chapter 2: Prepare the sample

In this chapter we will discuss the considerations needed before generating a single X-ray for the scan. The better prepared the sample is, either in knowing the intention for the scan or physically preparing the sample (positioning or otherwise) the better the data and results will be.

Scan requirements

When first considering how to scan a sample there are some choices that, if considered at the beginning of the process, makes getting ideal data far easier.

The theme of these choices is simple – what do you want to get from the data of the sample?

Resolution

Generally, the first decision you will make is – what resolution do you need?

Normally this will come from the size of the features you want to be able to inspect, or what resolution of measurements you want to make on the end data.

For example, you want to be able to inspect porosity within a part, with the porosities of interest being 30µm diameter or larger. Now, if we want to simply be able to resolve the pores (assuming they are separated by one pore diameter), we will need a scan resolution of 30µm.

However, if we want to be able to **measure** the porosity we will need at least 2-3 voxels, meaning we need a scan resolution of 10-15µm.

For this decision we discuss 'scan resolution' rather than just 'effective pixel size', since using a higher amount of power than optimal (and therefore a larger spot size than optimal) will decrease the scan resolution compared to the effective pixel size due to the larger geometric unsharpness.

Material(s)

The next decision is – what material are you interested in?

For single material samples this decision is already decided, but for multi-material samples this decision is very important, since:

- If you optimise for the lower density material, you may not adequately penetrate the higher density.
- If you optimise for the higher density material, you may not have enough contrast in the data to see details in the lower density.
- If you optimise for a balance, while you may get good detail for both, it wouldn't be the same as scanning the two materials separately.

This decision will also affect the artefacts (mainly X-ray penetration related) in the CT data since if you optimised for the lower density, then we can expect some predictable beam hardening artefacts along the longest pathlengths (although these can be mitigated via reconstruction tools such as Dual-Material CT).

Region of Interest

The next, and last consideration you will make for **every** scan is – what area of the sample are you interested in?

If you are interested in the whole sample then you need to ensure you can scan it at the required magnification position – if it doesn't with Circular CT, you may need to either:

- Perform multiple scans and merge the data together.
- Use field extension techniques such as Offset.CT or X.Tend to increase the size of the scan volume

If you are interested in only a section of the sample in the system, then as long as that section stays in the field of view for the entire scan (this may change with scanning mode e.g. X.Tend) you will be able to perform the scan successfully.

Noise level

The final consideration when setting up a CT scan is – what level of noise can you accept in the CT data? This will come with experience and is mostly dependent on the analysis you want to perform on the data.

For example, if you want to measure the geometry of a sample where the feature size is much larger than the scan resolution, then you can accept a higher level of noise since the scale of the noise (which is approximately on the scale of the voxel size) can be overcome via having good contrast in the data, or post-processing of the CT data (for example, using a median filter).

There are many situations where you will want to limit the level of noise in a scan are, for example:

- Performing porosity analysis where the pore size is less than or equal to the 2-3 voxel recommendation.
- Performing geometric measurements on the sample and requiring a higher level of accuracy
- Intending on visually inspecting features near the limit of the scan resolution
- Intending on inspecting the difference of very similar materials when the contrast of the scan is not optimal enough to easily define the two.

Sample position and support

Now we have considered what we want to achieve from the CT scan, we can start to get the sample into the system.

As mentioned at the start of this chapter, the physical positioning of the sample is very important.

This is because, with good sample positioning, we can:

- Limit (or eliminate) cone beam related artefacts generated from flat surfaces on the sample.
- Limit the length of the longest pathlength in the sample, which will allow us to obtain at least better contrast in the data since not as much filtration will be required, if not a quicker and/or less noisy scan.
- Limit any beam hardening artefacts being present in the region of interest, while positioning high density components away from the scanned volume (where possible).

For this section, we are going to show the effect of the sample position on the reconstructed data using a relatively simple sample – a coffee mug.

Sample with no support

The most basic support you can have for a sample is to be simply put it on the rotate stage. For some samples, where the area of interest on the sample is away from the stage, and the geometry is such that it doesn't cause artefacts (which will be explained later) this will be sufficient.

However, for most cases (such as this sample) it will not produce good results:

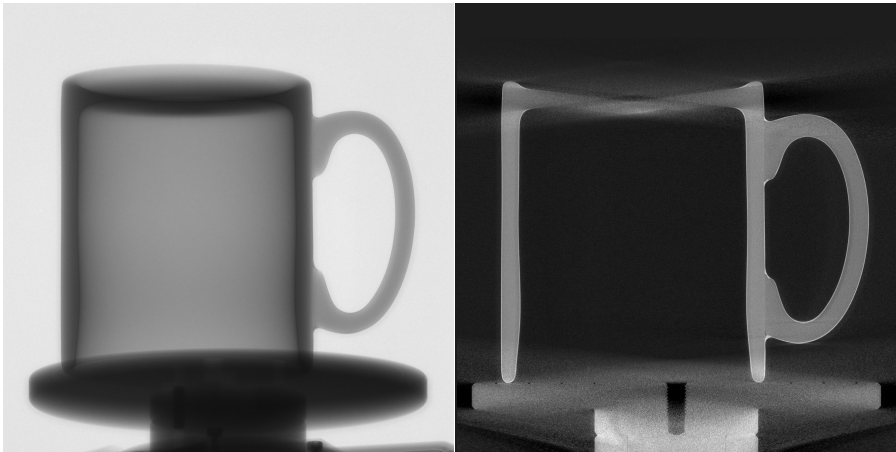


Figure 9. Scan of coffee mug simply placed on the rotate stage turntable. Left: Radiograph from scan. Right: slice from the obtained CT data.

The reasons this is not a good result are:

- We have an extreme difference in pathlengths within the sample – the base of the cup has a pathlength ~4 times greater than the next thickest pathlength (going through the handle). This is made worse by the base being horizontal since there is always this long pathlength in the radiographs. This long pathlength is seen by the darker region of the cup, which is a sign that part of the sample is under-penetrated.

- We have a lot of artefacts from the turntable (this will mostly be 'cone beam' artefacts, but there will also be an element of scattered x-rays as well due to the long pathlength of metal).
- We have almost completely lost the horizontal surface of the mug, which is due to 'cone beam' artefacts.

We have mentioned two of the most common artefacts you will see in CT data (with the other most common being 'beam hardening' artefacts), so it is important to **understand** what is causing them and how to avoid them.

Cone beam artefacts are a) lack of definition of horizontal surfaces, and b) the cloud of greyvalues coming diagonally from those surfaces.

They are caused by flat surfaces being horizontal in the field of view, since the detector never sees tangentially along that surface, so has no information on that surface.

These artefacts can be eliminated by either:

- Tilting the sample so that there are minimal surfaces horizontal in the field of view
- Performing an X.Tend CT scan, since the detector will 'see' horizontal surfaces from below, tangentially and from above to give good definition of all the surfaces regardless of their orientation

X-ray scattering is caused by interactions with the electrons in the sample and in general, is reduced by increasing the average X-ray beam energy. For more complete information please refer to chapter 1 (page 18).

Sample with horizontal support

The simplest way to improve the position of the sample is by placing it away from the turntable (in this case on a piece of foam, or similar rigid but low attenuation material).



Figure 10. Scan of coffee mug separated from the rotate stage by foam. Left: Radiograph from scan. Right: slice from the obtained CT data.

By doing this we avoid the scattering caused by the turntable; but still have the cone beam artefacts and large difference in pathlengths.

Sample with optimised support

To optimise the positioning of the sample, we now need to:

- Avoid parallel surfaces to the turntable (where possible)
- Minimise long pathlengths within the sample

We can satisfy both by angling the sample. The ideal angle would be 45° , which would optimise how much sample can fit in the field of view; but any angle from horizontal will:

- Reduce the cone beam artefacts
- Reduce the pathlengths through the base of the mug. There will always be an angle where the X-rays need to penetrate the full diameter of the mug base (i.e. when the base is aligned to the cone beam) but will be for a minimum amount of time.

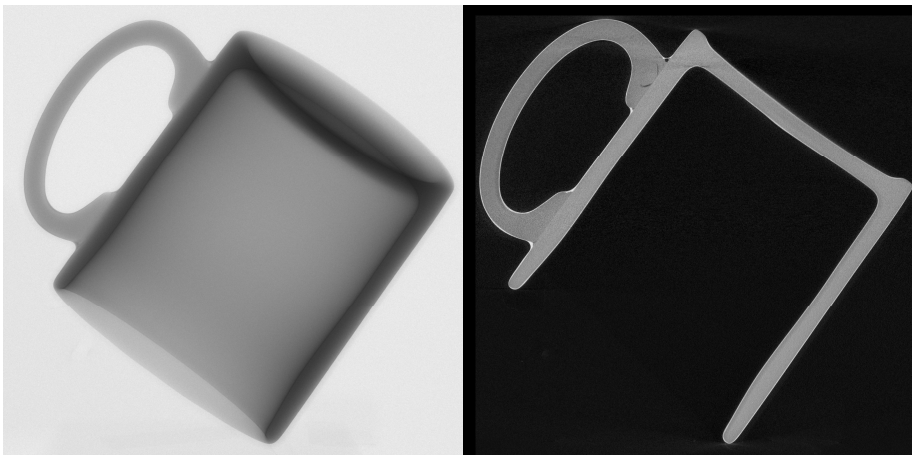


Figure 11. Scan of coffee mug separated from the rotate stage and placed at an angle by foam. Left: Radiograph from scan. Right: slice from the obtained CT data.

In the images, we can see that the only cone beam artefacts remaining are those on the curved sections where the surface is parallel to the turntable, which will be generated within this sample regardless of the orientation due to the shape of the mug – the best we can do is minimise the artefacts.

Chapter 3: Determine X-ray parameters

Now the sample is positioned within the system, we can now start to work on determining what conditions to use for the CT scan.

This section will guide you through the process of performing a **circular CT** scan.

Throughout this section we will use some specific terminology to discuss the radiographs. While these will be further explained later, it is useful to familiarize yourself with them before starting:

Detector Baseline	The average greyvalue of the radiograph when no X-rays are present. This gives a representation of the noise within the image.
Pathlength	A straight line between the focal spot of the X-ray source and a given pixel on the detector.
Darkest Pathlength	The most attenuating pathlength – travelling through the thickest and/or most dense part of the sample
Brightest Pathlength	The least attenuating pathlength (while still penetrating part of the sample). Generally, this is the thinnest wall of the sample being scanned.
Background	The average greyvalue of the radiograph when X-rays are present, but the sample is out of the field of view.
Effective pixel size	The amount of sample represented by each pixel on the detector. This is the main indication of the resolution of the CT scan.
Beam energy	The electric potential between the focus cup and the target within the X-ray source. The higher the value the more penetrating the X-ray beam produced.
Beam current	The current allowed to hit the target within the source. This doesn't change the characteristics of the X-rays produced.
Power	The multiplication of the beam energy and current. A higher power (beyond the best focus of the source) results in a lower focussed source focal spot.

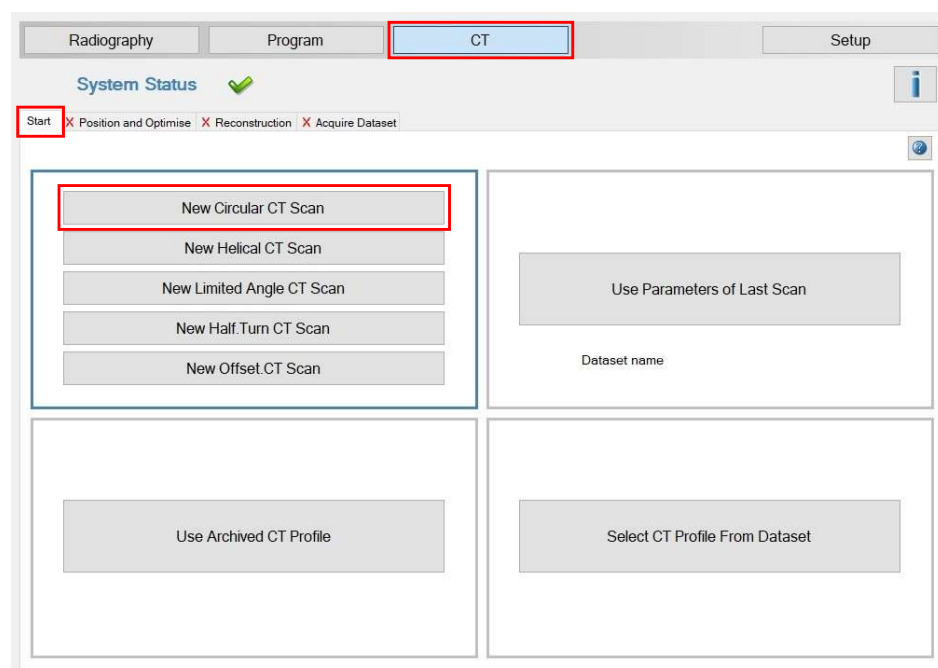
Step 1: Select CT scan mode

To start the process, we need to inform the system which kind of CT scan to perform.

To get to the below page we must select the 'CT' tab (which is always visible at the top centre of the Inspect-X workflow).

Select 'Start', then 'New Circular CT Scan'.

By pressing this button, the software will move you onto the 'Position and Optimise' tab.



Going back to the 'Start' tab after this point will leave the X-ray settings unchanged but will erase any information entered in custom fields such as 'Filter' or 'Dataset Name'.

Step 2: Initialise the axes

Before beginning any work, for a circular CT scan please ensure that:

x axis = 0 mm

tilt axis = 0 degrees

If these are not achieved, then the acquisition will not behave as expected since:

An incorrect x axis will reduce the size of the field of view

An incorrect tilt axis will cause doubling in the data. This cannot be recovered by usual means.

Step 3: Choose a starting point

To start the process, we now need to think back to our **scan requirements**. For this scan we have chosen:

Region of interest	The entire mug
Resolution	The best achievable while still scanning the entire mug
Material(s)	the mug is primarily ceramic with only a thin glaze coating, so we will not consider it a multiple material sample
Noise	Since no analysis is to be performed, the noise level isn't a significant factor for this scan. However, it will be kept to a minimum where possible.
Artefacts	Avoid artefacts from the longest pathlengths

Step 3a: Determine the resolution of the scan

The best starting point for the process is to determine the effective pixel size achieved while satisfying the region of interest.

For this step, the X-ray conditions are not essential, but a higher frame rate/(shorter exposure) is advised, so the image responds quickly with sample movement.

Since we are interested in the entire mug we have:

1. Put the centre of the mug (while in its angled jig) over the centre of the turntable
2. Adjust the y axis position such that the centre of the mug over the centre of the detector
3. Increase the magnification so that the sample fills the detector
4. Rotate the sample through a full rotation of the turntable. If the region of interest leaves the detector, then reduce the magnification such that it stays within it.

Note: any part of the sample leaving the field of view will not be scanned

It may take a couple of iterations to get the sample properly centred on the turntable. It can be easiest to rotate the sample, so it reaches the furthest left/right on the detector. You then **must physically move the sample on the turntable. Do not move the x-axis from 0mm**. Repeat this until you are satisfied with the centring of the sample through the full rotation of the turntable.

Step 3b: Determine the beam energy

The choice of what initial beam energy to use will greatly depend on the X-ray source being used, and what material thicknesses are being scanned.

In general, **the higher density materials will require a higher kV.**

The principles we will be covering mean that you can achieve an optimal scan regardless of what beam energy you start with, but the below are good starting ranges for common material groups, and the likely limiting factor(s) you will encounter in the scanning:

Material	Suggested initial beam energy	Challenges
Biological/Food	30 – 60kV	Contrast will be the focus to differentiate materials from each other and the background
Low density/ composites	60 – 120kV	Contrast , as above
Low attenuation metal pathlengths (low-density metal/thin high-density metal)	120 – 180kV	Penetration will start to become a primary concern to ensure enough signal is achieved through the longer path length. Contrast will be important if performing porosity analysis.
Thick low-density metal/high density metal	180 – 450kV	Penetration will be the primary concern
Multiple material samples	Compromise between materials	You need enough penetration to have signal through the higher density material, but enough contrast to differentiate the material(s) and the background

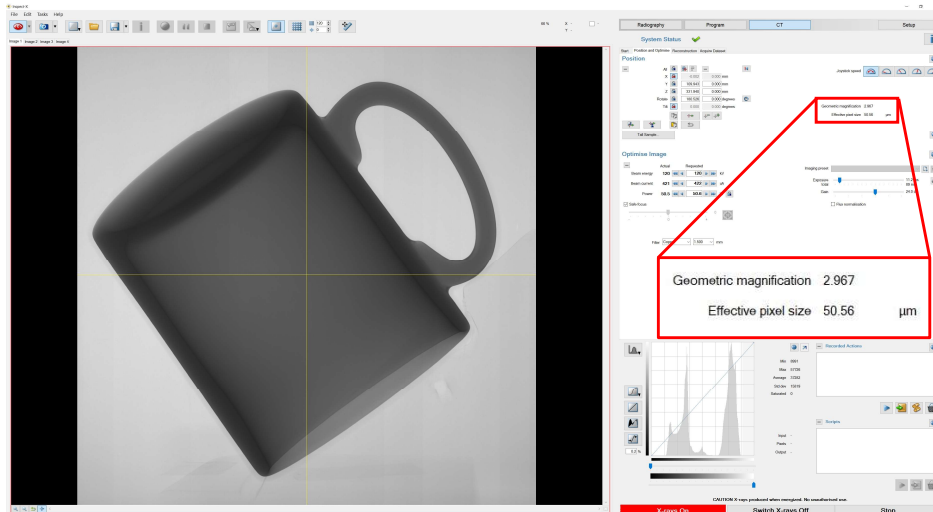
For the mug scan, 120kV was chosen as the starting point.

Step 3c: Determine the maximum optimal X-ray power

To limit the amount of unsharpness in the scan, we need to limit the amount of power (and therefore amount of source spot de-focussing) we use. This is explained on pages 15-16 & 21 (x-ray source focal spot size) and in further detail on page 20-21 (image resolution), and is based on the **effective pixel size**.

As general rules we can allow:

- 1W/ μm (effective pixel size) while using a static target
- 3W/ μm (effective pixel size) while using a rotating target



Based on the effective pixel size (given in the top right of the workflow), and that we are using a 225kV static reflection target the optimal maximum power we can use is 50.56W.

Step 3d: Set the imaging settings

For this step we want to get an initial image from the detector. It's not critical that this first image is perfect, but by applying some of our earlier decisions (which at this point will mostly be a consideration of what noise we want in the scan, and how long the scan can take), will aid us in this process.

For this scan, we are aiming for a compromise between noise and time. Therefore, a gain of 24dB was decided as a reasonable choice since 6-12dB would be a low noise scan, 18-24dB is an average scan, and 30-36dB is for rapid scans.

The frame rate can then be set, such that the panel is bright while not being saturated. Ideally the image will have greyscale values across the detector's range (0 - 65,536).

It's not uncommon to have the ideal point between two exposure settings. In this case you can:

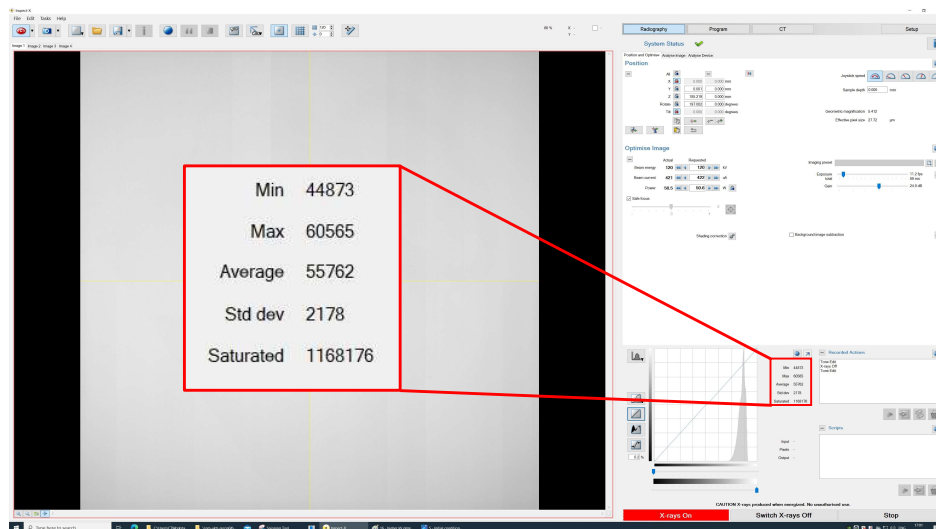
- **Optimise for speed.** To take this choice you will use the faster exposure setting and either leave the beam current as it was set (which will result in a darker image and therefore marginally higher noise).
- **Optimise for image quality.** To take this choice you will use the slower exposure setting and decrease the beam current such that there are no saturated pixels on the detector. This will keep the image using the whole of the detector's imaging range, and the decreased beam current (and therefore beam power) will also result in marginally sharper projections.

In the case of the mug scan, an exposure of 11.2fps (or 89ms) was found to be a good setting without having to make the above decision.

Step 4: Verify starting point

Before doing any in-depth checks on the sample/image we want to make sure that the values we have used – 120kV, no filtration, 50.56W, 24dB, 11.2fps – is able to be scanned.

What we need to check for this is whether any pixels on the detector are saturated when the sample is out of the field of view. We need no pixels to be saturated in this situation for the shading correction to function properly. If there are any then **step 3d** must be repeated for this clear-field-of-view position.



When checking this for our scan, we find that 1,168,176 pixels (~14.3%) of the detector are saturated. These will be in the central region of the detector since a) the sample was always covering this part of the detector, and b) the centre of the detector is marginally closer to the source focal point, so tends to be brighter than the edges of the detector.

For our re-consideration of step 3d, it was decided to optimise for quality so reduced the power to 47.3W.

Step 5: Check X-ray penetration through the sample

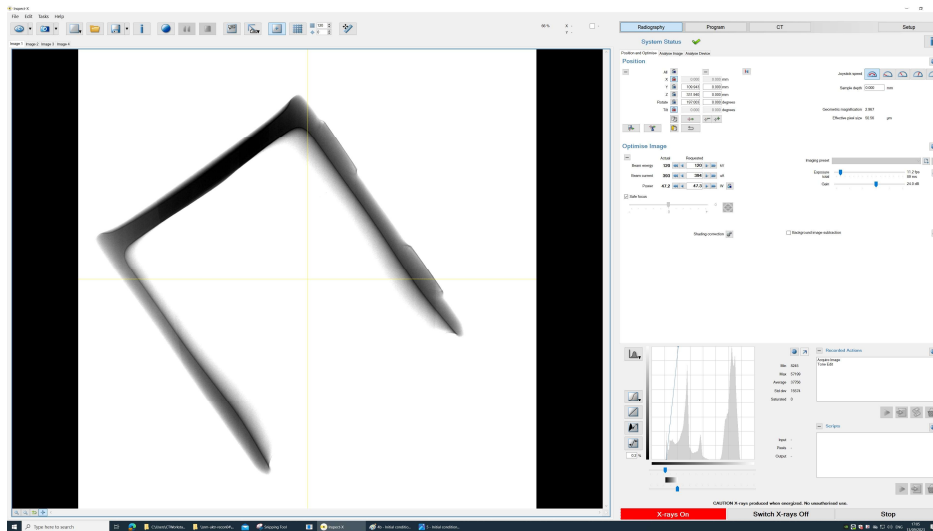
Now we have a reasonable starting point, we can now turn our attention to how the X-ray beam is interacting with the sample, and attempting to determine if the scan will yield a good result.

The next aspect of the scan to check is whether we are achieving enough signal through the darkest pathlengths.

Step 5a: Find the darkest greyvalue

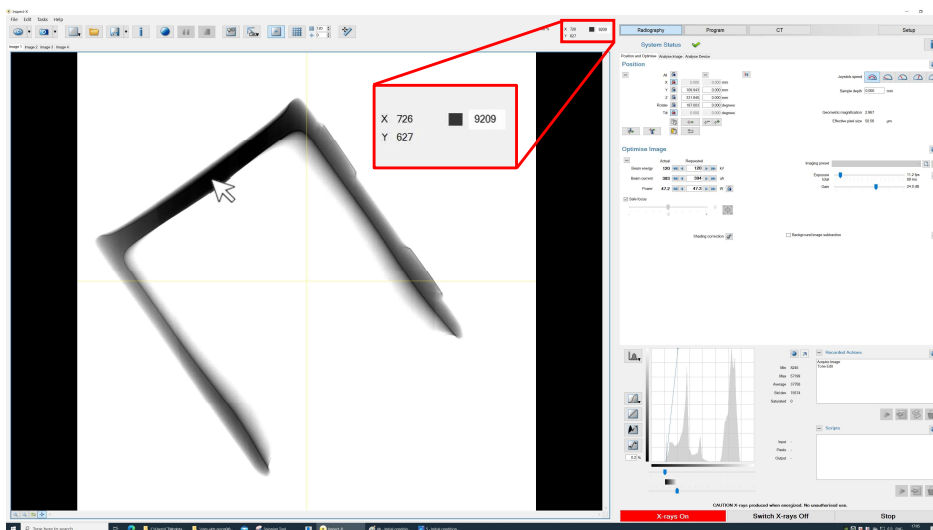
To do this we need to *find* what the worst pathlength(s) are. This will be where the thickest part of the sample is (parallel for the cone beam angle), or dense parts of the sample align.

The simplest way to do this without having to make complicated considerations of how the sample is positioned within the cone beam, is to adjust the tone curve to only show the darkest parts of the image (see below) and rotate the sample through 360 degrees. This should make it relatively simple to find where the darkest pathlength is.



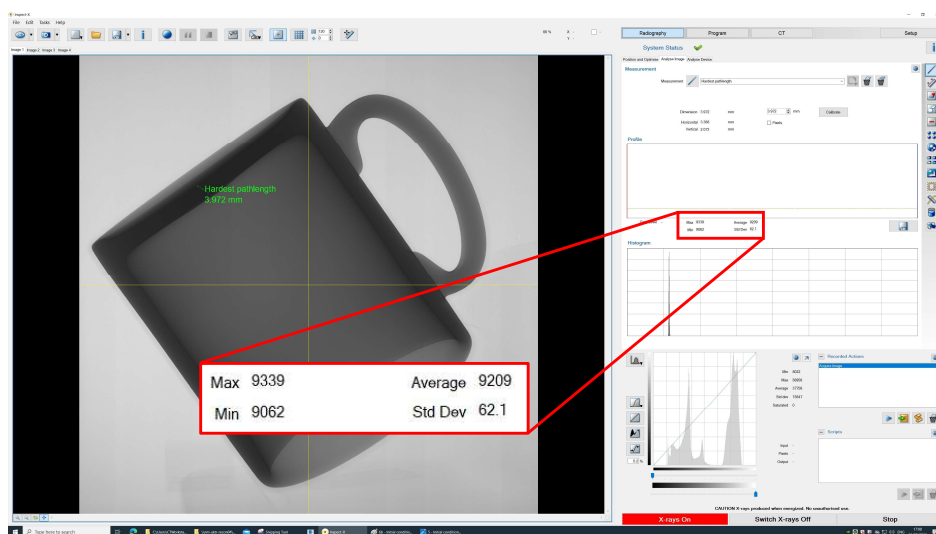
To find what the greyvalue is through this specific part of the image is, we have two methods:

- Hover the computer cursor over that part of the image. The greyvalue of where the cursor is positioned is displayed above the top-right of the image



- Acquire an image and use the measurement tool (Radiography tab > Analyse Image > Measurement > Linear measurement) to select an area of the darkest part of the image. This has the added benefit of not being affected by the noise of the image compared to the random selection of the mouse cursor since many pixels are being sampled.

This method was used for the discussions in this training guide. Use the 'average' value displayed in the region's statistics.

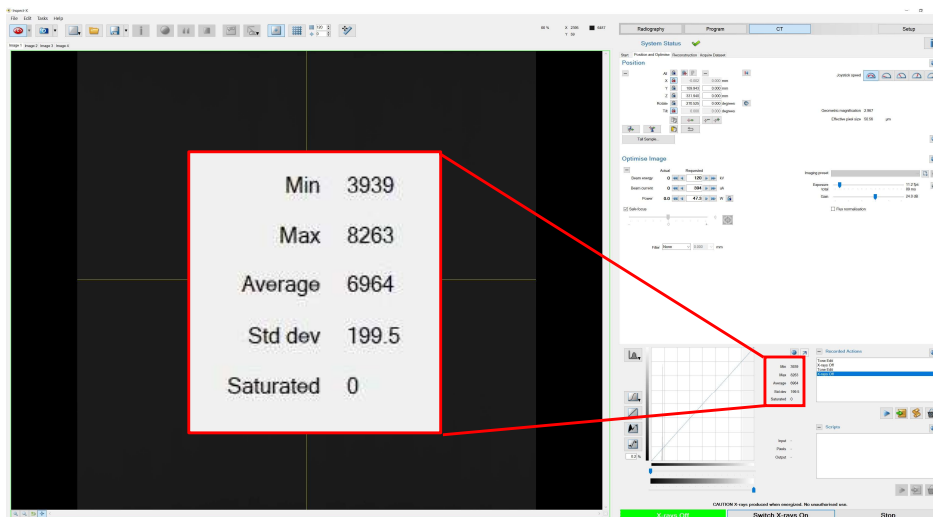


From the image we can determine that the darkest pathlength has a greyvalue 9209.

Step 5b: Validate the darkest greyvalue

Now that we have determined what the darkest pathlength is within the sample, we now need to check if this is good or bad.

To do this we compare the value we determined to the *detector baseline*. This is where imaging is live, but the X-ray beam has been turned off. After turning the X-ray beam off wait at least 10 seconds before taking a note of the image's average greyvalue. This is to allow the detector's pixels to fully discharge and therefore obtain a more accurate value of the baseline.



In this case we can see that the detector baseline is a greyvalue of 6964.

As a general recommendation, the darkest pathlength should satisfy:

$$\text{Darkest pathlength} \geq 1.5 * \text{Detector baseline}$$

Satisfying this condition means that there should be enough signal in the data to not cause significant artefacts from under-penetrating the sample. This is generally seen as an extreme noise grain (high and low greyvalues) in the under-penetrated directions.

In the case where a material is penetrated enough to not cause the major artefacts described above, there may still be an artificial greyvalue gradient with bright edges to the sample. This is **beam hardening** (see page 17) and is an indication that the material is not optimally penetrated. If this is observed in the data it can be corrected during reconstruction (see page ??), but for the optimal quality data it's recommended to have a consistent native dataset greyvalue resulting from more optimal scanning parameters.

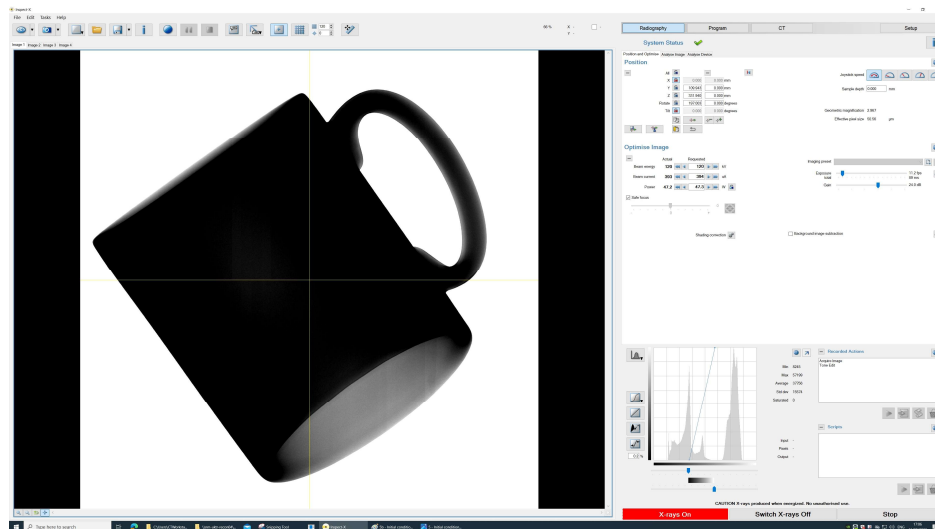
For a better quality scan it's recommended to have the darkest pathlength be greater than twice the detector baseline, but this is entirely sample dependent.

In the case of the mug scan, the darkest pathlength would ideally be at least 10446. Using the initial conditions, the darkest pathlength is only 32% above the baseline. Running this scan will still result in a reasonable scan but may have artefacts from the longer pathlengths.

Step 5c: Find the brightest pathlengths

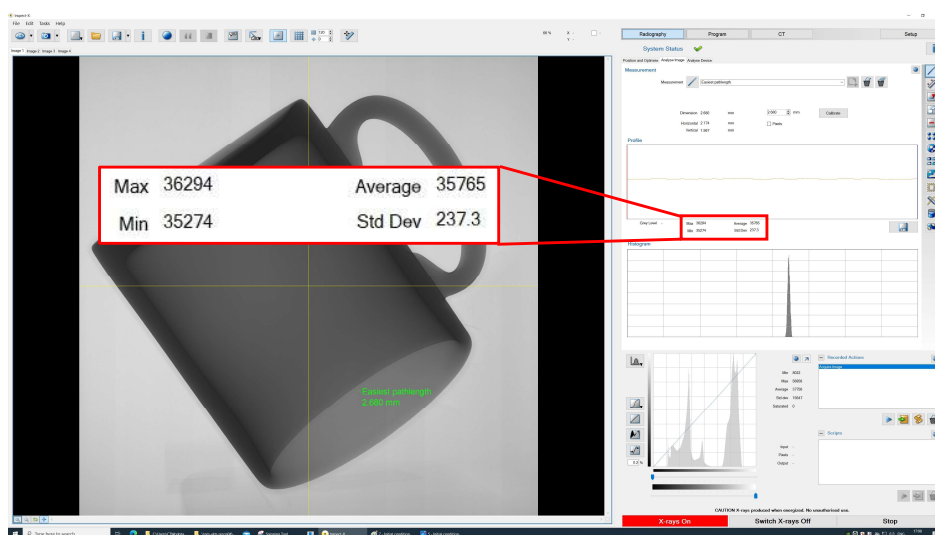
Now we have established the level of penetration we are achieving in the scan; we need to check that we aren't over-penetrating parts of the sample. This will have the same impact on the data as having too little penetration – not enough signal to overcome the noise of the scan.

Similar to finding the darkest pathlength, to find the brightest we will adjust the tone curve to highlight the brighter areas of the radiographs, for example:



In this case we find that the thinnest part of the sample is going through a single wall of the cup.

We can now use the same measurement tool to select an area of that thin region and find the average greyvalue.

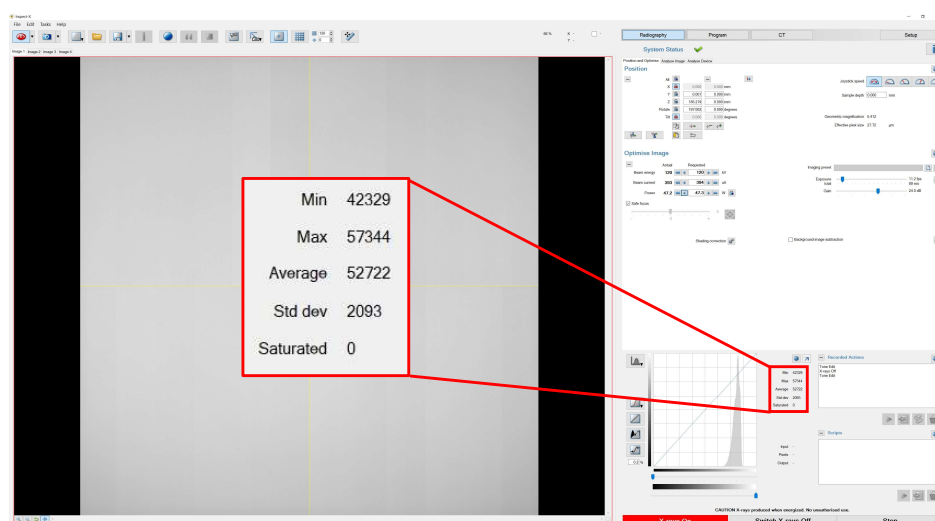


From the measurement we can determine that the easiest pathlength has a greyvalue of 35765.

Step 5d: Validate the easiest pathlength

To determine whether this value is good or not, we use a similar method to before, except we are comparing the obtained average to the background of the image.

For this comparison we simply need to move the sample out of the field of view, and check average of the image – in this case it's 52722.



As a general guideline it's recommended to have at least 5000 difference between the easiest pathlength and the background, so our value is sufficient with a difference of 16957.

Step 6: Adjust beam energy or source filtration, if required

The scan parameters we have determined have a reasonable amount of penetration with a large amount of contrast between the air and material.

Since we have a large difference between the air and material, generally we can increase the penetration by increasing the beam energy and/or filtration of the X-ray beam and still have a good amount of contrast.

That said, for the mug since the pathlength through the two walls and handle (the longest pathlength shorter than the base's) is ~16mm of material, while the pathlength through the base is ~80mm, we'll over-penetrate the easiest pathlength before reaching our "ideal" penetration through the base.

Given that this is the case, we will instead optimise for having the least artefacts while keeping a reasonable scan time.

This optimisation is simply going through steps 1-6 until results that meet your needs are achieved. It is important that as you change the filtration on the source, that you change the information entered in the 'Filter' text boxes/dropdowns. This way the scan can be repeated without unnecessary experimentation.

Below are the results of this optimisation before any reconstruction corrections are made:

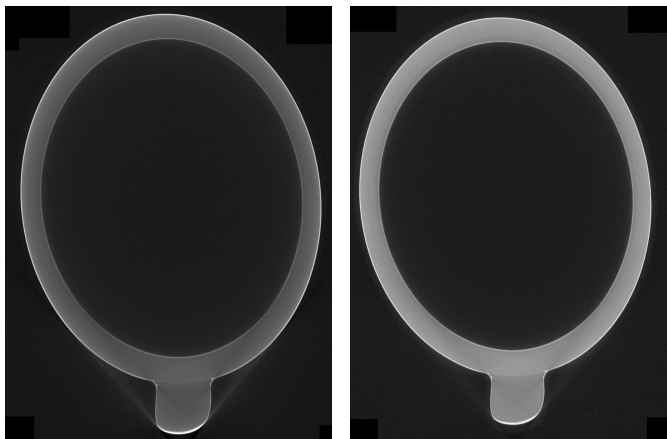


Figure 12. The scan with the initial settings (120kV beam energy with no filtration) is on the left side, and the final settings (215kV with 1.5mm Cu filter) is on the right side.

From the images we can see that the material in the optimised scan has a more consistent grey value with only a minor beam hardening gradient being present. This would make any analysis easier to run, and the more reliable since the raw data has less artefacts.

When optimising the settings it can be easier to think about trying to stretch the useful part of the data (the sample) across the widest range of the detector. The below is a graph showing how the detector was utilised for each scan we performed until we reached the final settings:

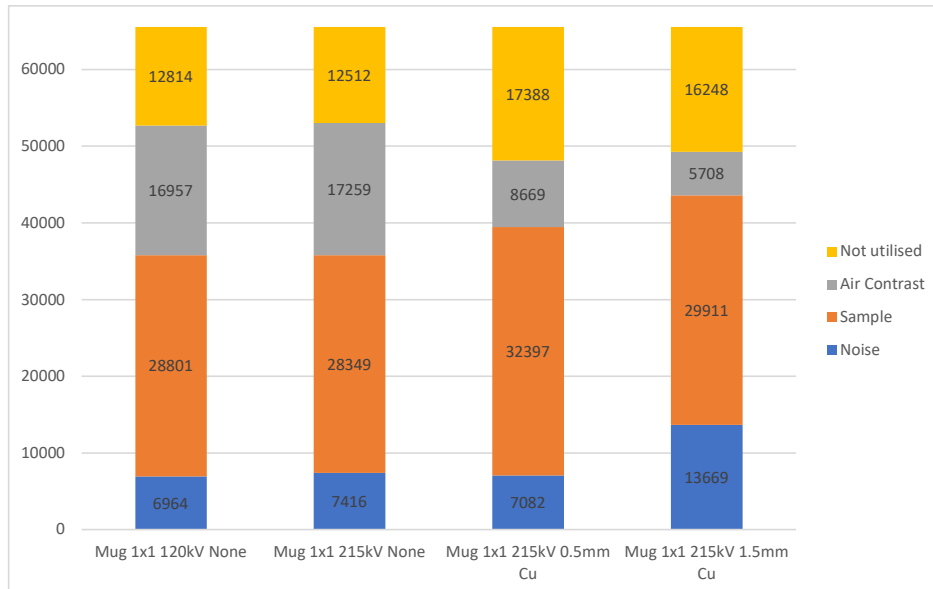


Figure 13. Bar graph showing detector utilisation. 'Noise' represents the level of greyvalue for the 'detector baseline'. 'Sample' is from the detector baseline to the greyvalue of the easiest pathlength. 'Air contrast' is the difference between the background greyvalue and the easiest pathlength. 'Not utilised' is the range of the detector not used in the scan (i.e. from the background greyvalue to 65,536)

As we increased the filtration, it increased the average energy of the X-ray beam, and therefore increased the penetration through the sample. This is easiest to see by the 'air contrast' value between the scans. A smaller value indicates that the beam is less affected by material as the easiest pathlength is becoming more similar to there being no material at all.

As the filtration was increased, the Signal-to-Noise level in the data decreased, which was to be expected as the filter removes the lower energy X-rays, so there is an overall flux decrease. This could easily be counteracted by performing a longer scan (which is explained in chapter 1)

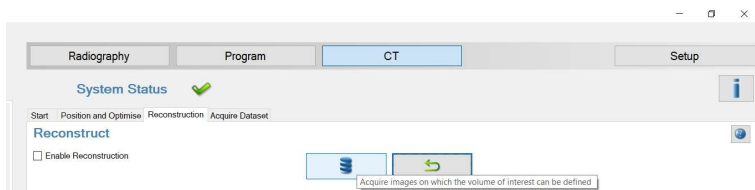
One point to note on the graph is that for the final scan, to preserve scan time the gain on the detector was increased from 18 to 24dB. This greatly increased the detector baseline, and therefore the noise in the final data. We also see this on the detector by the ratio of 'sample' to 'noise' being decreased.

Chapter 4: Set up the non-beam parameters

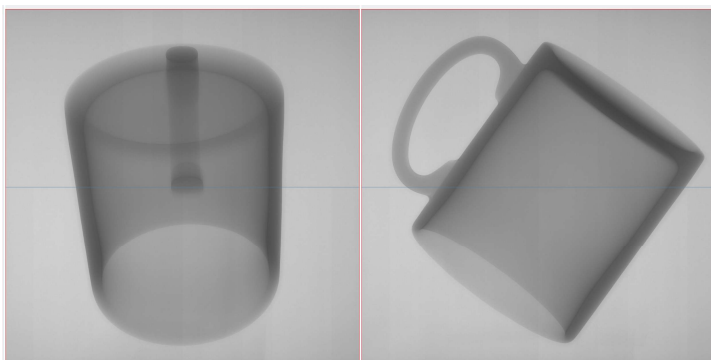
Now that we have determined what X-ray and imaging settings that are needed, we can proceed to the rest of the scan set up. We will now move to the 'Reconstruction' tab.

Step 1: Define the volume of interest

The first thing we will do is acquire two images of the sample that are a 90-degree rotation apart. We perform this action by pressing the highlighted button below.



Once the images are taken we will have the two images, with each one surrounded by a red box:



We can now adjust these red boxes by either clicking and dragging from within the images itself (image i), or by dragging in the red box sides (image ii):

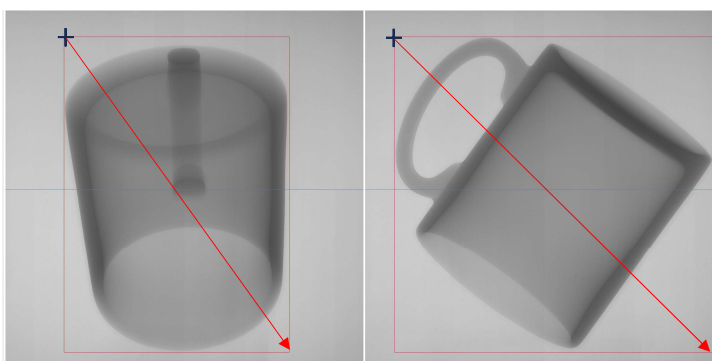


Image i. Generate a new red box by holding the cursor down from the top left corner and releasing it at the bottom right for each image. The direction of the drag is user choice (and not restricted by the software).

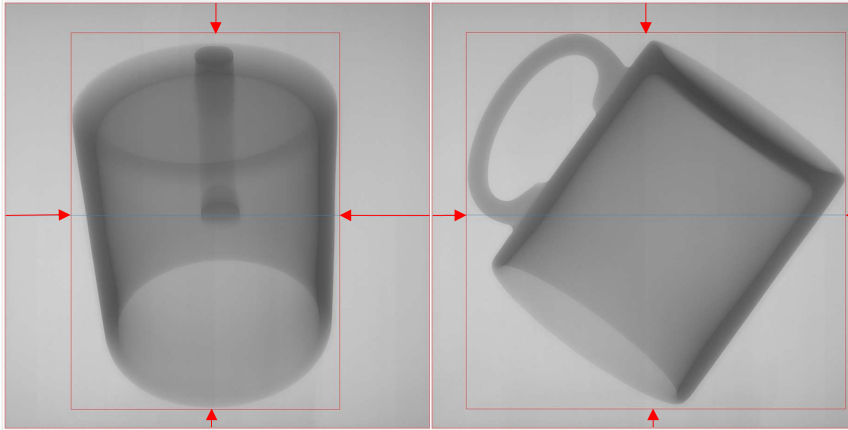


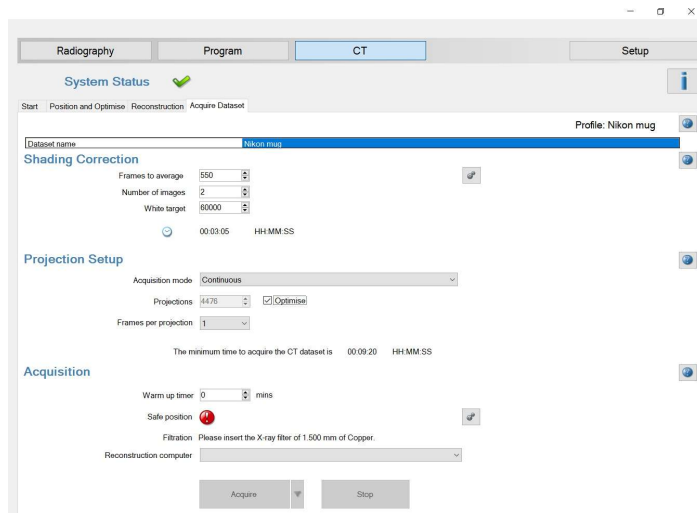
Image ii. Adjust the red box by dragging the sides to the desired position.

Unless the scan is being reconstructed automatically the optimal position of the box is such that it's touching the outer edges of the sample in the image.

Although we have set the height of the box, at this stage it's only the horizontal width of the box that is the critical feature we are concerned with as this will affect the number of projections we need to acquire.

Step 2: Give the dataset an identity

Now we have given the system the region of interest information, we can move onto the final tab – 'Acquire Dataset'.



This tab can be work from top to bottom, so we start by giving the dataset a name.

There is no software criteria for the name except there is a character limit, and some characters are not permitted. It's generally recommended that the name consists of:

[Operator identifier]_[Client/Project]_[Identification of the sample]

When the scan starts, the system will automatically append the scan start date and time, which means even if scans are given the same name they can still be identified.

Extra fields can be created for other information/meta data by adding additional lines within the properties [located at Properties > 3D CT > Dataset Information]. These extra lines will be found in the 'ctinfo' file.

Step 3: Shading correction

More information can be found in Chapter 1 for why we acquire a shading correction, but in short it is to make the detector uniform and therefore make it so that any difference on the radiographs is a result of the sample.

It's recommended that the number of images is set to 2, so that it takes a 0% power and a 100% power image. This is sufficient for most detectors and use cases.

The number of images to take depends on the exposure used in the scan parameters, but it should be such that noise is not a significant factor within the calibration images, which can be judged by the time taken to acquire the images.

The below table illustrates the recommended time to acquire a shading correction for each mode:

Scan Mode	Time (with auto filament control)	Time (without auto filament control)
Circular/Offset/X.Tend/ Limited Angle/Tilted	3 minutes 40 seconds	3 minutes 5 seconds
2-Position Panel scan	5 minutes	4 minutes 25 seconds
3-Position Panel scan	6 minutes 20 seconds	5 minutes 45 seconds
4-Position Panel scan	7 minutes 40 seconds	7 minutes 5 seconds

The times above derive from acquiring each correction image for 1 minute 20 seconds.

If auto filament control is enabled, it adds 35 seconds to the time estimate.

Also, between acquiring white (X-rays active) images and the single black (X-rays off) image there is a 25 second wait time which ensures the detector is discharged before acquiring the image.

Step 4: Projection setup

This section handles how many images, and how the system will acquire them for the scan.

Step 4i: Acquisition mode

There are up to three options for 'acquisition mode' – continuous, standard or minimise ring artefacts.

Continuous is normally the choice that will be made as it results in the fastest scans, since the sample continually rotates as the images are acquired. In theory this will add some minor movement blur to the radiographs, but in practice this is not seen in the final data unless very long exposures are used with very few projections (which is not recommended in any case).

Standard is the second most used, and functions by rotating to the requested angle and then stopping while acquiring the radiographs. This adds a large amount of time to short scans (2 or 3 times the scan time), but for longer exposure scans there isn't as significant difference (e.g. an additional 25% scan time).

This mode has an extra benefit for **X.Tend CT** where if the X-ray source 'trips' (experiences an over current and temporarily turns X-rays off) it can recover from the current position, while with 'continuous' scans it would restart the scan.

The final mode is 'minimise ring artefacts' and functions similarly to 'standard' rotation, but for each angle position it also acquires radiographs at slightly different x-axis positions. This means any misbehaving pixels will shift on the image and can then be removed during reconstruction.

Step 4ii: Projections

The optimal number of projections for a scan is based on the widest point a sample reaches on the detector. The optimal number is defined that no part of the region of interest moves more than one pixel between projections, meaning that wider sample will require more projections.

If **step 1** has been completed, by ticking the 'optimise' box the system will calculate the optimal number of projections based on the red box.

Projection Setup

Acquisition mode: Continuous

Projections: 4476 ☒ Optimise

Frames per projection: 1

The minimum time to acquire the CT dataset is 00:09:20 HH:MM:SS

The number of projections can be reduced by approximately a third without seeing major artefacts but is not recommended if scan quality is the priority. Increasing the number of projections can reduce the noise in the scan but does require more reconstruction time (and more disk space, which is a greater concern if it's required to keep all projections), and to see a noticeable effect would need to be increased by ~50% or more.

Step 4iii: Frames per projection

The default value for this setting is 1, where for each projection the detector will acquire a single radiograph and is always a 2-bit integer number (1, 2, 4, 8...).

Using 2 or more frames per projection will acquire multiple radiographs for each angle and then average them together to lower the noise in the data.

Increasing this setting will reduce the noise by a factor of $\frac{1}{\sqrt{\text{Number of frames}}}$ so to see a very large decrease in noise will increase the scan time considerably. If this is the aim, if it recommended to reduce the gain used in the 'position and optimise' tab before using the frames per projection setting.

Step 5: Acquisition

This final section handles actions around the scan itself.

Step 5i: Warm up timer

This setting would be used when starting a scan at high magnification with the X-ray source being not used for a duration before the set up.

The reason a 'cold' X-ray source can cause issues is that when the X-rays are generated most of the energy is transformed into heat, where this heat can cause the X-ray source to expand very slightly. This is a problem as the sources are generally mounted from the barrel end of the source, meaning that as the source heats up the X-ray focal spot can drift down, which causes the image to shift upwards on the detector and cause doubling in the final data.

This isn't needed at lower magnifications as this spot drift would be on the order of a pixel distance, or even sub-pixel distances.

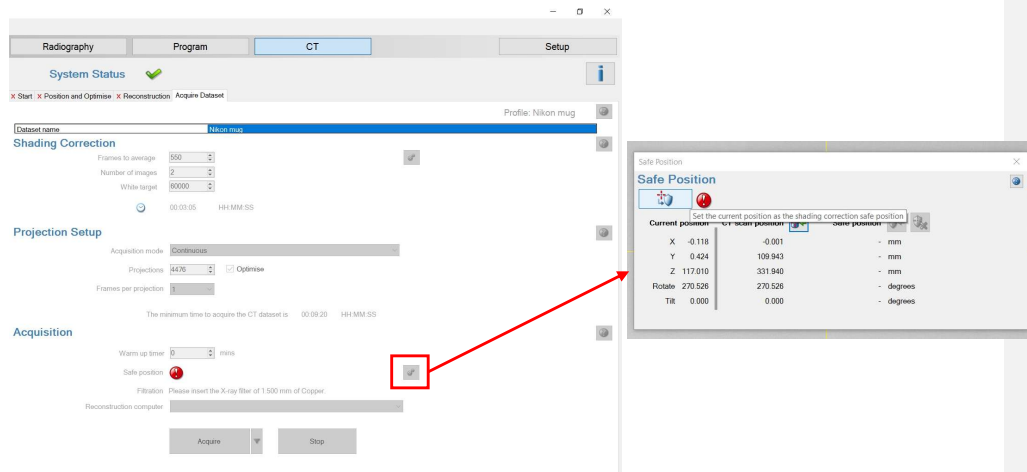
For a reflection source 2-3 minutes is usually sufficient, and for a transmission source a warm-up time of 5 minutes would be recommended since the source is capable of scanning at higher magnification positions.

Step 5ii: Safe position

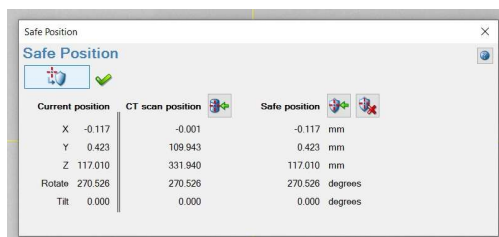
The safe position is a position where the sample is out of the field of view.

This can be set in two methods:

1. Move the sample out of the field of view on this tab and then proceed to acquire the scan (where the system assumes the position it is left at is the desired position).
2. Press the cog to the right (see below) and open the manual dialog



Using the manual dialog (above) allows us to set the safe position manually. To do this, we first move the sample out of the field of view, but then press the top left button which puts the current position as the safe position. If done correctly it should leave us with the below:

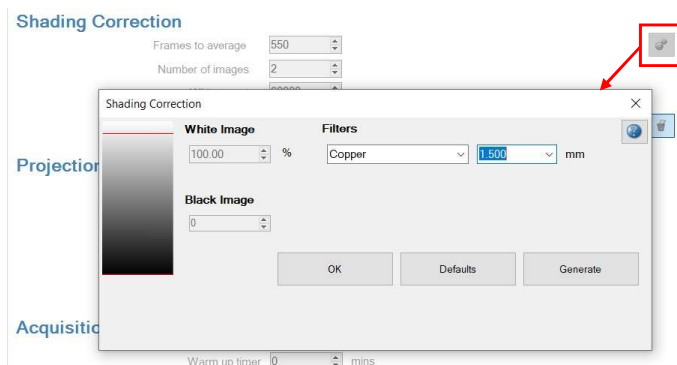


Step 5iii (optional): manual safe position

If there is no position where the sample is out of the field of view you can still acquire a shading correction.

Firstly, remove the sample from the system, and then move the manipulator out of the field of view.

Now press the cog button in the right of the 'shading correction' section. This will open the below dialog box:



To generate the shading correction, you simply need to press 'generate' after making sure the filter information entered is correct.

Once the process is complete (noted by a green tick appearing in the 'shading correction' section), the sample can be replaced and you can proceed to the rest of the scan.

Shading Correction



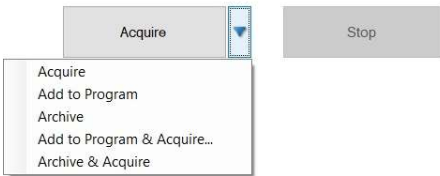
Step 5iv: Reconstruction computer

This is simply selecting which computer to send the dataset to.

Step 6: Acquire

The final step(s) of the scan set up is to choose what to do with the profile we have just generated. If just performing the scan is required, then press the 'Acquire' button and the scan process will start.

The other options are available via the dropdown which is the righthand side of the 'Acquire' button:

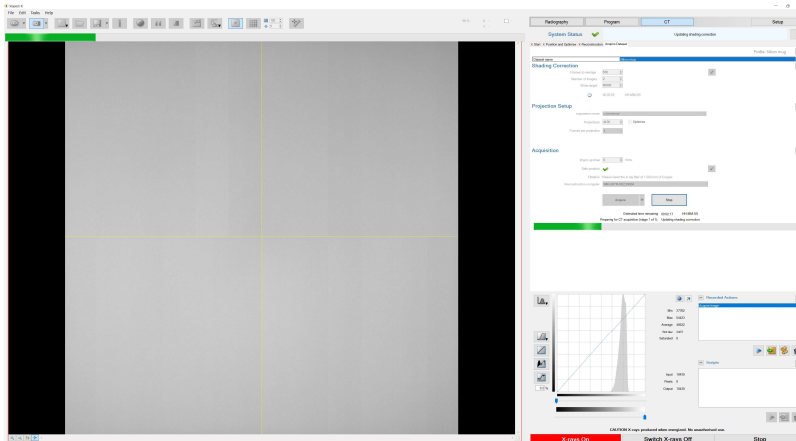


Acquire	starts the scan process
Add to Program	this allows the profile to be added to a CT batch (a queue of ctpfiles to be performed)
Archive	This saves the profile to the acquisition computer for easier retrieval. This is recommended if the scan profile will be used regularly. If an archived profile with the same name already exists, then a new version will be generated (the previous one is not overwritten).

Step 7: Scan started

If the acquire option is selected, then the scan process will start.

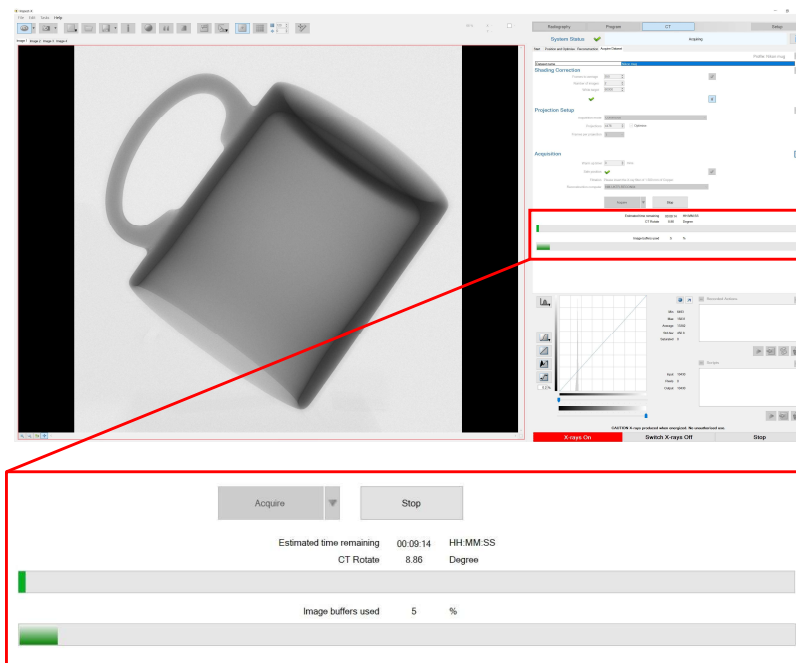
This will move the sample to the safe position and then generate a new shading correction if a valid one is not found. The progress of this step will be displayed such as below:



The progress of the specific image is shown as the green bar above the radiograph, while the overall process is shown on the right.

Once a shading correction is generated or found, the scan will start.

The sample starts rotating before the image acquisition begins, so that during the scan the sample has a consistent rotational speed. The progress of the scan is shown on the right of the workflow:





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